

Zudilin's Forms at Rational Bases: Proofs and Research Record

Erdős Problem #1049

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ABSTRACT

We specialise Zudilin's 2004 q -harmonic linear forms at rational bases. For coprime integers $a > b \geq 1$, put $\theta = \log b / \log a$. Then $F(a/b) = \sum_{n \geq 1} ((a/b)^n - 1)^{-1}$ is irrational if $\theta < \theta^* = 0.4056830213840605 \dots$, and

$$\mu_{\text{irr}}(F((a/b)^r)) \leq \frac{1 - \theta}{\theta^* - \theta} \quad (r \geq 1).$$

In particular, every positive integral power of $31/4$ is covered, with bound less than 301. The forms, permutation-group integrality, direction and constants are inherited from Zudilin. The proof here cancels before homogenising and establishes the cancelled degree and positive remainder needed for rational specialisation. For the different normalised Hankel family in Zudilin's 2016 construction, we prove at every rank

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6}, \quad [q^{\text{ord}_q V_N^*}] V_N^* = \frac{(N!)^2 (N+1)!}{2^N}.$$

A separate positive-measure argument gives fixed-real-base positivity and size estimates. These statements do not determine cyclotomic content or positivity of the coefficient sequence. We formulate those two questions for an explicit polynomial pencil and provide reproducible finite-rank calculations through rank eight. These certify coefficient-matrix positivity and real pencil roots below the Lambert value through that rank, without an all-rank moment representation. The remaining sections prove restrictions for specified clearing, rank-two and endpoint-jet constructions. They separate a divisible selector relation from a small nonzero integral form. No irrationality statement at $3/2$, or universal rational-base theorem, is obtained. The principal irrationality and formal-order results have source-applied Lean declarations; the ordinary proofs and their dependencies are given here.

* *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Reading map

The two principal arguments are independent. For rational specialisation, read Theorem 2.2, the cancelled-degree calculation and the positive remainder, then Theorem 2.3 and the irrationality-exponent bound. For the 2016 Hankel family, read Theorem 3.4 and the separate positive-measure argument. Section 3.2 then distinguishes coefficient moments from cyclotomic content. The degree-budget theorem concerns only its displayed base-uniform rank-two hypotheses. The endpoint-jet sections give necessary further non-vanishing and size obligations at $3/2$, not a proof at that base. For finite coefficient positivity and pencil roots, read the certificate statement and Proposition 3.6. The registry appendix is provenance material, not the theorem index.

1 Introduction

Let $t > 1$ be a rational number and let $\tau(n)$ count the divisors of n . Erdős Problem #1049 asks whether

$$F(t) = \sum_{n \geq 1} \frac{1}{t^n - 1} = \sum_{n \geq 1} \frac{\tau(n)}{t^n}$$

is irrational [2, p. 102].

The two forms agree by expanding $(t^n - 1)^{-1} = \sum_{k \geq 1} t^{-nk}$ and collecting equal exponents. Nonnegativity justifies the rearrangement, while $\tau(m) \leq m$ proves absolute convergence for every real $t > 1$. The coefficient of t^{-m} is the number of divisors of m . The question is a conjecture of Chowla; Erdős proved it for every integer $t \geq 2$ [1]. The catalogue snapshot cited here records that rational-base question and its attribution to Chowla [25]; it is not a guarantee that the bibliography is complete. Specialising Zudilin's 2004 construction [8] to rational bases, this note proves $F(a/b)$ irrational for every pair of coprime integers $a > b \geq 1$ with $\log b / \log a < \theta^* = 0.40568302138406054 \dots$, which settles the base $31/4$ and every power $(31/4)^r, r \geq 1$. This paper does not establish the universal conjecture over all rational $t > 1$. The cited literature already covers non-integral rational bases, including $7/2$ below; no claim of a first non-integral example is made.

Write $t = r/s$ in lowest terms with $r > s \geq 1$, so that $s = 1$ is exactly the integer case Erdős settled. Among reduced non-integral rational numbers greater than one, $3/2$ has the least naive height $H(r/s) = \max(r, s)$. It is the particular base whose remaining approximation obligations are examined here. A published height criterion of Bundschuh and Väänänen [6, Thm. 2, p. 177; hypotheses pp. 175–176] settles a family of rational bases restricted by a height condition; that family contains $7/2$ and does not contain $3/2$. Between the two lies the question this note is about: what exactly stops the integer-base argument from running at $3/2$?

Dependencies and conventions. The rational-base proof uses the 2004 coefficient polynomials; the Hankel proof uses the different 2016 normalised remainder sequence. The first is not a change of notation for the second. We use $q = p^{-1}$ for a fixed real base $p > 1$, and Gaussian polynomials $\begin{bmatrix} m \\ k \end{bmatrix}_p$ have base p . The approximation scale is n and the Hankel rank is N . Formal order as $q \rightarrow 0$, real asymptotics at fixed q , polynomial content in

$\mathbb{Q}[p]$, and specialised integer-row content are separate notions. A cited construction, an ordinary deduction, an exact finite computation and a kernel-checked declaration are identified where used, rather than collected under a single claim of verification. Empty products and determinants have value 1. All asymptotic estimates with a subscripted base hold with that base fixed. In the 2004 construction alone, $N = 15n$ denotes the cyclotomic cutoff; elsewhere N is a Hankel rank. The symbol $\mu = C_1/C_0$ is a construction parameter, whereas $\mu_{\text{irr}}(\xi)$ denotes the exponent of a particular irrational value.

Formal identity and historical integer-base results. The *Formal Conjectures* file for Problem #1049 contains the conjecture and integer-base theorem as sorry placeholders, but it also proves the Lambert identity between the two displayed series for rational t [28]. In the present $t > 1$ regime its `lambert_convergent` branch is an ordinary convergent-series proof; the same file's $|t| \leq 1$ branch instead uses Lean's convention that the `tsum` of a nonsummable series is zero. Thus it supplies genuine checked prior art for the identity, but no irrationality theorem. The present note formalises propositions about the clearing argument and likewise does not answer the conjecture.

Erdős's positive-integer theorem of 1948 [1] sits inside a larger integer-base literature.

Functional nonrationality is not value irrationality. At the level of functions, Rivin proves that if a sequence γ and its divisor-sum sequence are both eventually linearly recurrent, then γ is finitely supported [23, Theorem 1.1, p. 2; proof pp. 6–7]. His periodic-coefficient corollary therefore shows that

$$\sum_{n \geq 1} \frac{z^n}{1 - z^n}$$

is not a rational function [23, Corollary 6.4, p. 9]. This is an exact structural statement about the function underlying Problem #1049, but it gives no irrationality statement for a special value at $z = 1/t$: a nonrational function may take rational values at particular rational points.

Padé and orthogonal-polynomial constructions. In 1991 Borwein proved the irrationality of shifted series $\sum_{n \geq 1} (t^n + w)^{-1}$ at integer bases $t \geq 2$, for every nonzero rational w with $w \neq -t^m$ for all $m \geq 1$, by Padé approximation rather than by digit clearing; his estimates also show that these values are not Liouville numbers [3, Thm. 4, pp. 257–258]. In 2001 Van Assche recovered the integer-base irrationality and the bound $\mu(F(p)) \leq 2\pi^2/(\pi^2 - 2) = 2.50828 \dots$ using little q -Legendre Padé approximants [17, Thm. 1, p. 10; proof pp. 10–11]. His more general Theorem 3 proves irrationality of $\sum_{k \geq 1} (cp^k - 1)^{-1}$ for an integer $p > 1$ and fixed rational c away from the poles [17, Thm. 3, p. 14]; it does not cover a rational noninteger base t in $F(t)$, because the multiplier needed to write t^k over an integer base varies with k .

Related kernels and non-transfer of recurrences. The same Lambert value was already the target of Amdeberhan and Zeilberger's q -WZ construction [5]. Here $p > 1$ is the integer-base parameter and the little- q kernel uses $q = p^{-1}$. The two constructions share that bivariate little- q -Legendre Padé kernel, but Van Assche's diagonal does not

satisfy the Amdeberhan–Zeilberger scalar recurrence. Their Theorems 1 and 2 do prove, respectively, the irrationality of the non-alternating $h_p(1)$ and alternating q -logarithm values for their stated integer parameters, with reported irrationality measure 4.80; those are external integer-parameter results and do not transfer to the rational noninteger base $3/2$ studied here.

Amdeberhan–Zeilberger use the moving diagonal $P_n(p^{n+1} \mid p^{-1})$, whereas Van Assche uses $P_n(p^n \mid p^{-1})$. Van Assche also records, citing Borwein’s 1992 Lemma 2, the neighbouring evaluation $P_{n-1}(cp^{n+1} \mid p^{-1})$ [17, (23) and the following paragraph, p. 6]; see [4, Lemma 2, p. 143]. The shared kernel therefore does not license transfer of recurrence, endpoint, lattice, or valuation claims between the diagonals. Indeed, if $A_n(p) = P_n(p^n \mid p^{-1})$, exact substitution at $n = 0$ into the Amdeberhan–Zeilberger operator [5, Sec. 1.5, p. 2] leaves

$$-p(p-1)^2(p+1)(p^5 + 2p^4 + 2p^3 + 2p^2 + 2),$$

which is nonzero for every real $p > 1$. One nonzero residual is decisive for non-transfer of that recurrence.

Lean checks the [exact residual factorisation](#) and its [nonvanishing for \$p > 1\$](#) . These are finite statements at $n = 0$; they do not supply a recurrence for either diagonal. No recurrence, endpoint, lattice, or valuation statement for one diagonal is used for the other; the displayed exact residual is the sole claim made here about their incompatibility.

Coussement and Smet use little q -Jacobi polynomials for further Lambert series whose base parameter is the reciprocal of an integer [18, Thms. 1.2–1.3, p. 2; Sec. 2], and Koizumi and Yokoi identify one of their three-parameter Apéry-type approximations with the Coussement–Smet Padé approximants [19, Sec. 7, Prop. 7.1, p. 33].

There is a second distinction at rational base $3/2$. Over the checked range, the raw rational approximation errors decrease while the selected cleared integer forms grow from $n \geq 2$. This says that the chosen integerisation does not produce small linear forms; it does *not* say that the raw approximants fail to converge.

In 2013 Vandehey proved that $\sum_{n \geq 1} d(n)a_n/b^n$ is irrational whenever $b > 1$ is an integer and (a_n) ranges in a finite integer alphabet excluding zero; taking $a_n = (-1)^n$ completes the elementary digit method for integer bases $b \leq -2$ [20, Thm. 1.2, p. 2]. His companion theorem permits a finite alphabet of nonnegative integers containing zero, provided the coefficient sequence is not eventually zero [20, Thm. 1.1, p. 2]. All three retain an integer base. Luca and Tachiya’s periodic-coefficient theorem likewise proves irrationality at every integer base of absolute value greater than one; their full-support example strengthens this to joint linear independence for finite families of iterated divisor functions [21, Theorem A, p. 139; Example 1, p. 140]. Duverney and Tachiya refine the Chowla–Erdős method to linear independence results for Lambert series at an integer base q with $|q| > 1$ [22, Thms. 1.1–1.2, pp. 2–3]; their Corollary 1.1 generalises Vandehey’s Theorem 1.2 [22, p. 3]. Each of these results assumes an integer base. For the underlying small-integral-tail criterion, see also Duverney [33, Thm. 2, p. 4 of the author manuscript]. We use that elementary criterion only as methodological context, not his separate square-index linear-independence argument.

In 2004 Zudilin obtained the uniform bound $\mu(F(p)) \leq 2.46497868 \dots$ for every integer $p \notin \{0, \pm 1\}$, by way of Heine’s basic transform and a permutation group [8, Thm. 1,

p. 154; Secs. 4–5, pp. 159–162]. The ordinary-hypergeometric antecedent is Rhin and Viola’s S_5 action and twelve-coset denominator reduction for rational forms in $\zeta(2)$ [16, Sec. 3, pp. 38–42; Sec. 4, pp. 46–51].

Those sources motivate the permutation, denominator and polynomial- specialisation architecture of Section 5; the endpoint lemmas there are abstract and are not yet applied to either source’s actual coefficient family.

The neighbouring problem of Lambert subseries $\sum_{n \in A} (t^n - 1)^{-1}$ over a restricted index set A is treated by Kovač and Tao [24, Sec. 2.1.2 and Thm. 2.3, pp. 4–5].

The rational non-integer progress relevant here is instead the height criterion of Bundschuh and Väänänen [6, Thm. 2, p. 177], which gives the strongest explicit numerical threshold for this value among the previously published sources compared here; Theorem 2.2 below enlarges the region that criterion covers. Zudilin’s later Padé-and-Hankel treatment of the generalized q -logarithm also comments on non-integral rational bases: it remarks that its results can be given for non-integer $p = r/s$ with $|p| > 1$, under an assumption $\log |r| > c \log |s|$ for some computable $c > 0$ [9, Sec. 2, p. 4]. No value of c is computed there. That remark announces the shape of a rational-base extension for the generalized q -logarithm of that paper. Section 2 carries out such an extension for F itself: specialising the 2004 forms at a rational base and carrying out the denominator accounting shows that $c = \mu = 2.464978683574975 \dots$, the 2004 bound quoted above, is admissible, and gives the resulting region and its application to $31/4$.

A third published rational-base region for this same series is Duverney’s Théorème 2, whose hypothesis is $\log |s| / \log |r| < \frac{1}{3}(1 - 3/\pi^2) = 0.2320 \dots$ [13, Théorème 2, p. 174]; his Théorème 1 for general numerators is weaker still. Both cutoffs lie strictly inside the Bundschuh and Väänänen region. Matala-aho, Väänänen and Zudilin’s combined treatment of q -logarithms keeps the integer hypothesis $p = 1/q \in \mathbb{Z} \setminus \{0, \pm 1\}$ throughout, and states in print that its methods do not sharpen the q -harmonic case of [8] [14, abstract p. 879; introduction p. 880]. So none of the printed regions for F compared here reaches $\log 4 / \log 31$.

Erdős’s integer-base argument, in outline. For an integer $b \geq 2$, Erdős rewrites $F(b) = \sum_{n \geq 1} \tau(n)b^{-n}$. A Chinese-remainder construction forces arbitrarily long blocks in which the divisor coefficients have the powers of b needed to make the corresponding base- b digits zero. Explicit bounds control the middle and far tails, while positivity proves that the expansion does not terminate. The resulting base- b expansion has arbitrarily long zero blocks without being eventually zero and is therefore irrational [1, pp. 63–66].

The direct cut-and-clear attempt studied here. A more naive attempt is to suppose $F(b) = p/q$, cut at N , and multiply the remaining identity by qb^N . This does not by itself trap a positive integer below 1: the first uncleared tail contribution is $q\tau(N+1)/b$. The sections below isolate additional corridor hypotheses under which a bounded-window version of this attempt would work, and then show why those hypotheses fail at $3/2$.

Write $\beta = r/s$ for the base and $c(n)$ for the coefficient of β^{-n} , so that $c = \tau$ in the case at hand. The term $c(n)\beta^{-n}$ is $c(n)s^n/r^n$. Clearing the power of r leaves the numerator

factor s^n in place. That factor is invisible when $s = 1$ and grows geometrically when $s \geq 2$. At $3/2$ it is 2^n .

Terminology. The linear-form method uses polynomial pairs before specialisation and integer pairs afterwards. We reserve *row* for an integer pair (U, V) . Its *integer scalar content* is $\gcd(|U|, |V|)$; a row is *primitive* when this number is 1, and dividing by it is *primitive normalisation*. A polynomial pair may instead have a polynomial common factor in $\mathbb{Z}[X]$; that is a different operation and is not called row content here. The *exterior determinant* of two integer rows is $U_n V_m - U_m V_n$, the determinant of the 2×2 matrix they form. Two quantities attached to that determinant are compared throughout: an integer dividing it, which is a local gain, and its absolute value, which is an Archimedean cost; we call that comparison the *local-to-Archimedean balance*.

The *endpoints* of a coefficient polynomial, taken relative to the declared width W of Section 5, are its constant coefficient and its coefficient at W ; we call these the *constant endpoint* and the *top endpoint*, so the top endpoint is the coefficient at W and not the leading coefficient unless the two agree. A *unit endpoint* is one equal to ± 1 , the units of $\mathbb{Z}$; Theorem 5.1 is the reason only these two coefficients decide divisibility by 3 and by 2 after specialisation at $(3, 2)$. Here a *jet* means a residue of a specialised coefficient modulo a prime power, not a derivative jet: the bottom jet is its residue modulo 3^R and the top jet its residue modulo 2^S , and R and S are the bottom and top *depths*. A jet vanishes exactly when the prime power in question divides the specialised coefficient.

The shortfall at $3/2$. The elementary method and the linear-form method are both examined below. For the height argument, the exact bracket $2^{64} < 3^{41} < 2^{65}$ gives a gap greater than $3/13$ across the whole admissible rectangular exponent cone and fixes the $39/41$ denominator-charge comparison. Neither the source scalar factor nor that factor together with the universally forced first-order residual border reaches the required charge. These comparisons leave higher residual valuations, determinant cancellation, and other integral models untouched. For the additive argument, the same bracket replaces the generic $4R + 2S$ threshold by $130T + 2S$ at depth $R = 41T$ and gives an exact first-row failure at $T = 1$.

For the coordinatewise clearing scheme the leftover at each step is the forcing term of an exact recurrence, of size at least 2^{N+1} whenever $s \geq 2$ and the scaling constant B and the coefficient $c(N+1)$ are at least 1 (Theorem 7.2), and the scheme itself is excluded at $3/2$ for every shift $N \geq 1$ and every cleared window of width $K \geq 1$ (Theorem 6.5).

For the linear-form constructions we examine two possible sources of 2-adic and 3-adic gain. Integer scalar content is exactly neutral, since it scales the exterior determinant and its absolute height by the same factor (Theorem 4.1), while unit endpoints keep both 2 and 3 out of any common divisor of the two specialised evaluations (Theorem 5.1 and Proposition 5.3). Corollary 5.5 states the two scoped exclusions together; neither is claimed to be necessary for every linear-form proof. Separately, the scalar parameter margin is negative under the assumed source inequality (Theorem 5.12).

One candidate pursued here is additive: an integer relation among rows that cancels the endpoint jets. Theorem 5.7 shows that a nonzero relation with coefficients in $\{-1, 0, 1\}$ cancelling all four jets exists whenever the bottom depth is positive and the

number of coefficient pairs is at least $4R + 2S$. It does not show that the resulting combination has a nonzero polynomial pair or a nonzero remainder. Problem 10.1 gives a precise sufficient specification for that particular candidate architecture, not a necessary condition for solving Problem #1049.

Sections 8 and 9 record two external methods and what each leaves unproved at $3/2$.

Sharpness. The following sharpness claims concern the stated models, not all approximation families. Two questions of scope are worth isolating. The corridor bound of Theorem 6.2 is exponential on the left and linear on the right, so for a fixed numerator and any $s \geq 2$ a corridor can survive only for bounded $N + K$; the base $3/2$ is the case in which the crossing has already happened at the smallest admissible window, which is why the exclusion there holds for all $N \geq 1$ and $K \geq 1$ with no further restriction. The height criterion of [6] is restricted by a height condition satisfied at $7/2$ and not at $3/2$, so the two bases are separated by that criterion rather than by a universal obstruction. The exponent 65 in $3^{41} < 2^{65}$ cannot be replaced by 64. The separate direct comparison $2^{129} < 3^{82}$ then proves that at bottom depth 41 the least number of input rows satisfying the ambient-cardinality inequality is exactly $130 + 2S$: $129 + 2S$ does not satisfy it. That is a statement about the counting test, and a family with fewer rows may still have a collision.

For $R = 41T$ with $T > 1$, the displayed $130T + 2S$ bound is uniform and sufficient and is not always the least such number.

Scope of the present results. The results below concern the displayed propositions and the supplied formal-source snapshot. The paper does not prove irrationality at $3/2$. An exact finite calculation, a conditional formal interface and a source-applied all-rank declaration have different scopes; the appendix records their separate scopes and dependencies.

Results and boundary. Theorems 2.2 and 2.3 specialise the construction of [8]. Its Lemma 7, (23), p. 161 is a polynomial integrality statement, prior to (24), p. 162, where an integer base is substituted. The direction and constants are also those of that source. The cancelled-degree and positive-remainder arguments needed here are proved in Section 2; the reciprocal-interval summation follows [7, Lemma 1, p. 466]. The supplied public snapshot contains the source-applied declarations `PaperR17.rational_base_region` and `PaperR17.thirtyone_four_powers`, rather than just conditional interfaces or finite height comparisons. The formal-source appendix specifies their evidence boundary. The cited Bundschuh–Väänänen analytic criterion at $7/2$ and the functional nonrationality and Mahler theorems remain external inputs. At $3/2$, an actual noncollapsed family with sufficiently small nonzero divided remainder is still missing.

Structure. Section 2 states and proves the region theorem, the base $31/4$ and the degree-budget cap on the mechanism that produces them. Section 3 derives the height and charge exclusions from the sharp $41/65$ power certificate. Section 4 proves that integer scalar content is neutral for the local-to-Archimedean balance.

Section 5 proves the endpoint congruences at $(3, 2)$, deduces from them and from Section 4 that neither integer scalar content nor a common divisor of the two specialised evaluations supplies the targeted endpoint gain, gives the four-jet collision count and its conditional Bézout–Plücker compression, and records one further exclusion on Zudilin’s scalar parameters. Sections 6 and 7 return to the elementary clearing scheme and record what the residue s^n costs there, first as an exclusion and then as an exact recurrence with a lower bound on the surviving term. Sections 8 and 9 record what is and is not formalised of two external routes, together with the separate 81/200 logarithmic comparison. Section 10 separates kernel escape from asymptotic adequacy and states the remaining obligations. Linked phrases identify formal-source declarations. The source map records the supplied snapshot and the historical direct-link revisions separately; a link is not itself evidence of a fresh Lean build.

Keywords. irrationality; Lambert series; rational base; Padé approximation; Lean 4.
MSC 2020. 11J72 (primary); 11J82, 68V20 (secondary).

2 The region $b^\mu < a$ and the base 31/4

This section states the note’s leading results. Fix the direction

$$(\alpha_0, \alpha_1, \alpha_2; \beta) = (14, 12, 14; 27)$$

from Zudilin [8, Sec. 5, pp. 161–162], and put

$$C_1 = (\alpha_0 + \alpha_1 + \alpha_2)\beta - \frac{1}{2}(\alpha_1^2 + \alpha_2^2 + \beta^2) = \frac{1091}{2}, \quad C_0 = 266 - \frac{3}{\pi^2}(225 - J),$$

$$J = \sum_{i=1}^{13} (\psi_1(u_i) - \psi_1(v_i)), \quad \psi_1(x) = \sum_{k \geq 0} \frac{1}{(k+x)^2},$$

the thirteen demi-intervals $[u_i, v_i)$ being $[1/14, 1/12)$, $[1/7, 1/6)$, $[3/14, 1/4)$, $[2/7, 1/3)$, $[5/14, 2/5)$, $[3/7, 7/15)$, $[1/2, 8/15)$, $[4/7, 3/5)$, $[9/14, 2/3)$, $[5/7, 11/15)$, $[11/14, 4/5)$, $[6/7, 13/15)$, $[13/14, 14/15)$. The constants are defined by these exact expressions, rather than by their decimal displays. The rational certificate accompanying this revision gives

$$\begin{aligned} 77.94318445520543 < J < 77.94318449473569, \\ 221.30008815898543 < C_0 < 221.30008817100119, \\ 0.40568302137302 < \theta^* := C_0/C_1 < 0.40568302139506. \end{aligned}$$

Put $\mu = 1/\theta^*$. The direction, intervals and constants, including $C_1/C_0 = 2.46497868 \dots$, occur in [8, p. 162]; that ratio is the integer-base exponent bound in its Theorem 1, not a new optimised constant. The displayed interval endpoints are rounded outwards from rational bounds; no digits beyond their certified accuracy are required.

The proof below needs J in its combinatorial form as well. Put

$$\omega(\xi) = \max\{0, \lfloor 14\xi \rfloor + \lfloor 13\xi \rfloor - \lfloor 12\xi \rfloor - \lfloor 15\xi \rfloor, 2\lfloor 14\xi \rfloor - \lfloor 13\xi \rfloor - \lfloor 15\xi \rfloor\},$$

the weight that [8, p. 162] attaches to the six-tuple $c = (13, 14, 12, 14, 15, 13)$ of the fixed direction; it gives the exponents $\nu_l = \omega(n/l)$ of the source’s (22) and enters its limit (26).

Zudilin lists the support of ω in the same place; the next lemma verifies that list by exact evaluation.

Lemma 2.1 (the weight is an indicator). *On $[0, 1)$ the function ω takes only the values 0 and 1, and $\omega = 1$ exactly on the union of the thirteen demi-intervals listed above.*

Proof. Each of the three expressions inside the maximum is constant on every interval between consecutive elements of $\{k/c : c \in \{12, 13, 14, 15\}, 0 \leq k \leq c\}$, since those are the only points at which one of the four floor functions changes. There are 48 such intervals in $[0, 1)$; evaluating ω at the midpoint of each gives the value 1 on the thirteen listed demi-intervals and 0 elsewhere, and the thirteen listed intervals are unions of consecutive cells. The evaluation is exact rational arithmetic on finitely many points. $\square$

Writing $1_{[u,v)}$ for indicators, Lemma 2.1 and termwise integration of the positive series $-\psi'_1(\xi) = \sum_{k \geq 0} 2(k + \xi)^{-3}$ give the two forms of J used below,

$$J = \int_0^1 \omega(\xi) \sum_{k \geq 0} \frac{2}{(k + \xi)^3} d\xi = \sum_{i=1}^{13} (\psi_1(u_i) - \psi_1(v_i)),$$

the first being the integral $\int_0^1 \omega d(-\psi_1)$ of [8, Lemma 2, p. 155, and (26), p. 162]. Since ω vanishes near 0, both are finite.

Theorem 2.2 (rational-base region). *Let $a > b \geq 1$ be coprime integers with*

$$b^\mu < a, \quad \text{equivalently} \quad \frac{\log b}{\log a} < \theta^* = 0.40568302138406054 \dots$$

Then $F(a/b) = \sum_{m \geq 1} ((a/b)^m - 1)^{-1}$ is irrational.

The proof occupies Sections 2.1–2.4. Its integrality input is the polynomial conclusion of Zudilin’s Lemma 7 together with the coefficientwise argument that proves it [8, pp. 156–161], and it uses his direction and constants. The cyclotomic limits of Section 2.3 are his Lemmas 1–2, proved there by the method of [7, Lemma 1, p. 466] for the degree count. The rational specialisation and the accompanying degree and positive-remainder arguments are supplied here. This describes the division of work in the proof; it is not an exhaustive priority claim about rational specialisations.

2.1 The source forms as polynomial identities

The strict threshold is essential to this proof: at $\log b / \log a = \theta^*$ the leading quadratic balance is zero, and the error estimates stated here do not decide its sign.

Fix $n \geq 1$, write x for an indeterminate or a real number greater than 1 according to context, and put $q = x^{-1}$. Set

$$\begin{aligned} a_0 &= 14n + 1, & a_1 &= 12n + 1, & a_2 &= 14n + 1, \\ \beta_n &= 27n + 2, & N &= 15n, & M_n &= 266n^2 + 34n + 1. \end{aligned}$$

The symbol β_n is the parameter b of [8, Sec. 3, p. 155], the exponent in the lower parameter q^b of the Heine series (13) on p. 157; the letter b continues to denote the denominator of the base, and M_n is the integer of that paper's (16) on p. 157. With $(z; q)_m = \prod_{j=0}^{m-1} (1 - zq^j)$, the source's summand is

$$H_n(x) = \sum_{t \geq 0} \frac{(q^{t+1}; q)_{a_1-1}}{(q; q)_{a_1-1}} \frac{(q; q)_{\beta_n-a_2-1}}{(q^{a_2+t}; q)_{\beta_n-a_2}} q^{a_0 t}, \quad (1)$$

whose last denominator has the length $\beta_n - a_2$ given by the gamma expression (7) and the residues (8) of [8, p. 156], and [8, (8)–(11), pp. 156–157] gives $H_n = A_n F - B_n$ with $A_n, B_n \in \mathbb{Q}(x)$. The two coefficients can be written down. Let $\begin{bmatrix} m \\ r \end{bmatrix}_X$ be the Gaussian binomial polynomial, of degree $r(m-r)$ and with nonnegative coefficients summing to $\binom{m}{r}$, and for $a_2 \leq k \leq \beta_n - 1$ put

$$c_{n,k}(X) = (-1)^{a_1+a_2+k+1} X^{e_{n,k}} \begin{bmatrix} k-1 \\ a_1-1 \end{bmatrix}_X \begin{bmatrix} \beta_n-a_2-1 \\ \beta_n-k-1 \end{bmatrix}_X, \quad (2)$$

$$e_{n,k} = \frac{a_1(a_1-1)}{2} - \frac{(\beta_n-a_2)(\beta_n-a_2-1)}{2} + \frac{(\beta_n-k)(\beta_n-k-1)}{2}.$$

Then

$$A_n(X) = \sum_{k=a_2}^{\beta_n-1} c_{n,k}(X) X^{a_0 k}, \quad (3)$$

$$B_n(X) = \sum_{k=a_2}^{\beta_n-1} c_{n,k}(X) X^{a_0 k} \left(\sum_{l=1}^{k-a_1} \frac{1}{X^l - 1} + \sum_{j=1}^{a_0-1} \frac{X^{-j(k-a_1)}}{X^j - 1} \right). \quad (4)$$

Put

$$D_N(X) = \prod_{l=1}^N \Phi_l(X), \quad v_{n,l} = \omega(n/l), \quad \Omega_n(X) = \prod_{l=2}^N \Phi_l(X)^{v_{n,l}}, \quad (5)$$

with Φ_l the l th cyclotomic polynomial; by Lemma 2.1 every $v_{n,l}$ is 0 or 1, and $v_{n,l} = 0$ for $l > 15n$ because the intervals all begin at or above $1/14$.

Zudilin's Lemma 7 [8, p. 161, (23)] applies at the parameters above: the tuple is $c = n \cdot (13, 14, 12, 14, 15, 13)$ with maximum $m(c) = 15n$ and difference $s(c) = n > 0$, and the source's (14) on p. 157 holds because $12n+1 \leq 14n+1$ and $26n+2 \leq 27n+2 \leq 28n+2$. Its conclusion is the inclusion

$$U_n := X^{-M_n} \frac{D_N}{\Omega_n} A_n \in \mathbb{Z}[X], \quad V_n := X^{-M_n} \frac{D_N}{\Omega_n} B_n \in \mathbb{Z}[X]. \quad (6)$$

Two points about that citation matter. First, the conclusion used is the one in $\mathbb{Z}[X]$, not integrality at integer arguments: an integer-valued polynomial such as $X(X-1)/2$ shows that the two statements differ. The source's proof supplies the polynomial form. Its Lemma 4 (pp. 157–159) clears the two rational coefficients separately by Gaussian-binomial identities in $\mathbb{Z}[X]$, its Lemma 5 (p. 160) is the stability of the normalised series under the order-twelve group generated by the parameter map that Heine's transform (1) induces and the exchange of a_1 and a_2 , and Lemma 7 (p. 161) removes the

cyclotomic factors by polynomial divisibility, Ω_n being monic. All three precede the source's evaluation at an integer p in its (24) on p. 162, and all three remain valid with an indeterminate. Second, the pair extracted from the source's module inclusion is unique, because F is not a rational function of X : as $h \downarrow 0$, splitting the original Lambert sum $F(e^h) = \sum_{m \geq 1} (e^{hm} - 1)^{-1}$ at $L = \lfloor 1/h \rfloor$ gives an initial sum $h^{-1} \sum_{m \leq L} m^{-1} + O(L)$, since $y^{-1} - 1 \leq (e^y - 1)^{-1} \leq y^{-1}$ for $0 < y \leq 1$. The remaining sum is at most $e^{-h(L+1)} / ((1 - e^{-1})(1 - e^{-h})) = O(h^{-1})$. Hence $F(e^h) = h^{-1} \log(1/h) + O(h^{-1})$, so $(X - 1)F(X) \rightarrow \infty$ and $(X - 1)^2 F(X) \rightarrow 0$ as $X \downarrow 1$, which no rational function does. Hence the coefficients of F and of 1 in an identity over $\mathbb{Q}(X)$ are determined, and (6) identifies them. This is a statement about F as a function and says nothing about any individual value.

2.2 Exact degrees

Let $d_{n,k} = \deg(c_{n,k}(X)X^{a_0 k})$. The Gaussian-binomial degree formula gives

$$d_{n,k} = e_{n,k} + a_0 k + (a_1 - 1)(k - a_1) + (\beta_n - k - 1)(k - a_2) = \frac{-k^2 + 80kn + 3k - 340n^2 - 26n}{2},$$

$$d_{n,k+1} - d_{n,k} = 40n + 1 - k > 0 \quad (a_2 \leq k \leq \beta_n - 2),$$

the inequality because $k \leq \beta_n - 2 = 27n$ and $27n < 40n + 1$. The summand of top degree is therefore the one at $k = \beta_n - 1$ alone, no cancellation occurs there, and

$$\begin{aligned} K_n &:= \deg A_n = d_{n,\beta_n-1} = \frac{1091n^2 + 81n + 2}{2}, \\ W_n &:= \deg U_n = K_n - M_n + \sum_{l \leq N} \varphi(l) - \sum_{2 \leq l \leq N} v_{n,l} \varphi(l), \end{aligned} \tag{7}$$

the second equality because $\deg D_N = \sum_{l \leq N} \varphi(l)$ and $\deg \Omega_n = \sum_{2 \leq l \leq N} v_{n,l} \varphi(l)$, and because U_n is a polynomial by (6), so subtracting M_n from the degree of $D_N A_n / \Omega_n$ is legitimate. In particular $U_n \neq 0$.

Four arithmetic checks in this step are represented by the supplied Lean declarations: $2M_n$ against the source's (16) is [one declaration](#), the degree step $d_{n,k+1} - d_{n,k} = 40n + 1 - k$ is [a second](#), its positivity on the whole range $a_2 \leq k \leq \beta_n - 2$ is [a third](#), and the top value $2K_n = 1091n^2 + 81n + 2$ is [a fourth](#). They decide the arithmetic of this step and no analytic step anywhere in the note.

For the second coefficient, fix n and let $x \rightarrow \infty$. Every factor in (1) tends to 1 and the geometric factor $q^{a_0 t}$ makes the sum converge, so $H_n(x) = O(1)$, while $F(x) = \sum_m \tau(m)x^{-m} = O(x^{-1})$. From $B_n = A_n F - H_n$ and $\deg A_n = K_n$ it follows that $B_n(x) = O(x^{K_n-1})$; multiplying by $x^{-M_n} D_N(x) / \Omega_n(x)$ and using integrality (6) gives

$$\deg V_n \leq W_n - 1. \tag{8}$$

One common homogenising width W_n therefore serves both coefficients. The degree calculation fixes n and varies x ; the decay calculation fixes x and varies n . Polynomial integrality was established before the first calculation, so an analytic bound at infinity legitimately bounds the polynomial degree rather than merely a rational-function growth rate.

2.3 The cyclotomic limit

The limits proved in this subsection are those of [8, Lemmas 1–2, p. 155]. The proof splits the condition $\{n/l\} \in [u, v)$ into reciprocal blocks and applies the summatory totient estimate on each, as in the proof of [7, Lemma 1, p. 466], and adds an explicit truncation estimate.

Write $\Sigma_n = \sum_{2 \leq l \leq N} \nu_{n,l} \varphi(l)$. The elementary summatory estimate is $\sum_{l \leq y} \varphi(l) = 3\pi^{-2}y^2 + O(y \log(2y))$. Fix one demi-interval $[u, v)$ of Lemma 2.1. The condition $\{n/l\} \in [u, v)$ holds exactly on the blocks

$$\frac{n}{k+v} < l \leq \frac{n}{k+u}, \quad k = 0, 1, 2, \dots,$$

so the summatory estimate gives, for each fixed k , $n^{-2} \sum_{l \text{ in block } k} \varphi(l) \rightarrow 3\pi^{-2}((k+u)^{-2} - (k+v)^{-2})$. The blocks with $k \geq L$ consist of integers $l \leq n/(L+u)$, and $\varphi(l) \leq l$, so their total is at most $\frac{1}{2}(n/(L+u))^2 + O(n)$ and their normalised contribution is $O(L^{-2})$ uniformly in n . Taking $n \rightarrow \infty$ first and then $L \rightarrow \infty$ therefore justifies the infinite block sum, and summing the thirteen demi-intervals gives $n^{-2}\Sigma_n \rightarrow 3J/\pi^2$. With $\sum_{l \leq 15n} \varphi(l) = 3\pi^{-2}225n^2 + O(n \log n)$, equation (7) yields

$$\frac{K_n}{n^2} \rightarrow C_1, \quad \frac{K_n - W_n}{n^2} = \frac{M_n - \sum_{l \leq N} \varphi(l) + \Sigma_n}{n^2} \rightarrow 266 - \frac{3}{\pi^2}(225 - J) = C_0, \quad (9)$$

and hence $W_n/n^2 \rightarrow C_1 - C_0$. These are the rational-base degree limits built on Zudilin's cyclotomic limits [8, Lemmas 1–2, p. 155; (26), p. 162].

For the Archimedean side, fix a real $x > 1$. Put $B(x) = \sum_{d \geq 1} -\log(1 - x^{-d})/d < \infty$. The Möbius product formula and a divisor reindexing give

$$\sum_{l \leq N} |\log \Phi_l(x) - \varphi(l) \log x| \leq \sum_{d \leq N} [N/d] [-\log(1 - x^{-d})] \leq NB(x).$$

Since $\nu_{n,l} \in \{0, 1\}$, this proves the sharper bound

$$\log \frac{D_N(x)}{\Omega_n(x)} = \left(\sum_{l \leq N} \varphi(l) - \Sigma_n \right) \log x + O_x(N). \quad (10)$$

Here $N = 15n$, so the error is $O_x(n) = o(n^2)$. The bracket equals $W_n - K_n + M_n$ by (7). No uniform constant as $x \downarrow 1$ is asserted.

2.4 Positive integral forms and the contradiction

Fix the base $x = a/b > 1$ and write $P_q = (q; q)_\infty > 0$. Each of the four finite q -Pochhammer products in (1) is a product of factors $1 - q^j$ with $j \geq 1$, hence lies in $[P_q, 1]$, so each of the two ratios lies in $[P_q, P_q^{-1}]$, every summand is positive, and

$$P_q^2 \leq H_n(x) \leq \frac{P_q^{-2}}{1 - q^{a_0}} \leq \frac{P_q^{-2}}{1 - q},$$

whence $H_n(x) > 0$ and $\log H_n(x) = O_x(1)$ uniformly in n . Cyclotomic polynomials are positive on $(1, \infty)$, so by (6)

$$\Lambda_n(x) := U_n(x)F(x) - V_n(x) = x^{-M_n} \frac{D_N(x)}{\Omega_n(x)} H_n(x) > 0.$$

By (7) and (8) both polynomial degrees are at most W_n , so

$$\widehat{U}_n = b^{W_n} U_n(a/b), \quad \widehat{V}_n = b^{W_n} V_n(a/b)$$

are integers, and $\widehat{\Lambda}_n := b^{W_n} \Lambda_n(a/b) = \widehat{U}_n F(a/b) - \widehat{V}_n$ is positive and lies in $\mathbb{Z}F(a/b) + \mathbb{Z}$. This is the rational-specialisation denominator-clearing step. The inherited polynomial integrality and the preceding cyclotomic cancellation are separate arithmetic inputs. Combining the size estimates,

$$\begin{aligned} \log \widehat{\Lambda}_n &= W_n \log b - M_n \log x + \left(\sum_{l \leq N} \varphi(l) - \Sigma_n \right) \log x + O_x(n) \\ &= K_n \log b - (K_n - W_n) \log a + o(n^2), \end{aligned}$$

the second equality by $\log x = \log a - \log b$ and (7). With (9) this proves

$$\lim_{n \rightarrow \infty} \frac{\log \widehat{\Lambda}_n}{n^2} = C_1 \log b - C_0 \log a. \quad (11)$$

Proof of Theorem 2.2. The hypothesis $\log b / \log a < \theta^* = C_0 / C_1$ makes the right side of (11) negative, so $\widehat{\Lambda}_n > 0$ and $\widehat{\Lambda}_n \rightarrow 0$. Suppose $F(a/b) = P/Q$ with integers P and $Q \geq 1$. Then $Q\widehat{\Lambda}_n = P\widehat{U}_n - Q\widehat{V}_n$ is a positive integer for every n and tends to 0, which is impossible. $\square$

Above the threshold (11) says that these same homogenised forms grow like e^{cn^2} with $c > 0$, and at equality the limit decides nothing. Both statements are about the displayed family.

The hypothesis $b \geq 1$ admits $b = 1$, where the statement reduces to the known integer-base theorem. Coprimality is a restriction on the pairs treated and is used nowhere in the argument. Negative bases are excluded, because Step 4 is a positivity argument.

Theorem 2.3 (the base $31/4$ outside the Bundschuh–Väänänen region). *$F(31/4)$ is irrational, and so is $F((31/4)^r)$ for every integer $r \geq 1$. Here*

$$\frac{\log 4}{\log 31} = 0.4036981731641997 \dots < \frac{81}{200} < \theta^*,$$

while $4^\mu = 30.483515 \dots < 31 < 4^{\mu_{\text{BV}}} = 32.369642 \dots$ with $\mu_{\text{BV}} = 2\pi^2/(\pi^2 - 2) = 2.508284761994 \dots$, so $31/4$ lies outside the region $\log b / \log a < 1/2 - 1/\pi^2 = 0.3986788163576622 \dots$ of [6, Thm. 2, p. 177] and inside the region of Theorem 2.2.

Proof. The comparison $\log 4 / \log 31 < 81/200$ is the integer certificate $4^{200} < 31^{81}$, checked as [the power certificate](#), and the membership it yields is [checked](#); the ratio $\log b / \log a$ is invariant under $(a, b) \mapsto (a^r, b^r)$, which gives the [power family](#), and $(31^r, 4^r)$

are coprime. The exclusion from the earlier region is [checked](#). That exclusion also has a two-line rational certificate: $31^2 < 4^5$ gives $\log 4 / \log 31 > 2/5$, and $\pi^2 < 10$ gives $1/2 - 1/\pi^2 < 2/5$, so

$$\frac{1}{2} - \frac{1}{\pi^2} < \frac{2}{5} < \frac{\log 4}{\log 31} < \frac{81}{200} < \theta^*.$$

The remaining comparison $81/200 < \theta^*$ is proved in Section 2.5. Theorem 2.2 then applies. $\square$

Corollary 2.4 (an irrationality measure uniform over powers). *For coprime $a > b \geq 1$ with $\theta = \log b / \log a < \theta^*$ and every integer $r \geq 1$,*

$$\mu_{\text{irr}}(F((a/b)^r)) \leq \frac{1 - \theta}{\theta^* - \theta}.$$

Here $\mu_{\text{irr}}(\xi)$ is the supremum of the exponents ν for which $|\xi - p/q| < q^{-\nu}$ has infinitely many reduced rational solutions. In particular, $\mu_{\text{irr}}(F((31/4)^r)) < 301$ for every $r \geq 1$.

Proof. The raw source coefficient is a sum of $O(n)$ Laurent monomials times two Gaussian polynomials, each of coefficient sum at most 2^{27n+2} . Its coefficient norm is $\exp(O(n))$, and every exponent is at most K_n . The same normalising multiplier used above therefore gives

$$|U_n(x)| \leq x^{W_n} \exp(O_x(n)) \quad (x > 1 \text{ fixed}).$$

Set $\xi = F(a/b)$, $Q_n = b^{W_n} U_n(a/b)$ and $P_n = b^{W_n} V_n(a/b)$. With

$$\alpha = (C_1 - C_0) \log a, \quad \tau = C_0 \log a - C_1 \log b > 0,$$

we have $\log(Q_n \xi - P_n) = -\tau n^2 + o(n^2)$ and $|Q_n| \leq \exp(\alpha n^2 + o(n^2))$.

For integers A, B, p, q with $q > 0$, if $L = A\xi - B$ and $2q|L| \leq 1$, then

$$|L| \leq |A| |\xi - p/q|.$$

When $Ap - Bq = 0$ this is equality. Otherwise the nonzero integer $Ap - Bq$ gives $1/q \leq |L| + |A| |\xi - p/q|$, which proves the inequality. For a fixed $\eta \in (0, \tau)$ choose $n = \lceil \sqrt{\log(2q)/(\tau - \eta)} \rceil$ and apply it to (Q_n, P_n) . The two-sided remainder estimate yields

$$|\xi - p/q| \geq q^{-(\alpha + \tau + 2\eta)/(\tau - \eta) - o(1)}.$$

Let $\eta \downarrow 0$. The resulting bound is $1 + \alpha/\tau = (1 - \theta)/(\theta^* - \theta)$. Taking a common power multiplies α, τ by r , leaving this quotient unchanged; the constants in the approximation inequality may depend on r . For $31/4$, the exact rational interval calculation accompanying this revision gives

$$0.40568302137302 < \theta^* < 0.40568302139506,$$

$$0.40369817316419 < \frac{\log 4}{\log 31} < 0.40369817316420.$$

For each trigamma difference, sum $k = 0, \dots, 255$ exactly and bound the remaining decreasing positive summand $f_{u,v}(x) = (x+u)^{-2} - (x+v)^{-2}$ by its integral from 256 to

infinity and that integral plus $f_{u,v}$ (256). Bound π with Machin's identity and alternating arctangent series, and logarithms with the positive arctanh series and a geometric tail bound. Outward rational interval operations then place $(1 - \rho)/(\theta^* - \rho)$, where $\rho = \log 4 / \log 31$, between 300.4269130 and 300.4269164, hence below 301. The script `computations/intervals.py` records the rational endpoints; this enclosure is exact computer arithmetic, not a new Lean theorem. $\square$

2.5 The rational bracket around θ^*

The constant θ^* is defined by a trigamma series, so a decidable membership test needs explicit rational bounds on it. Both are finite rational arithmetic, and neither uses a decimal expansion.

For the lower bound, keep only the $k = 0$ term of each of the thirteen differences $\psi_1(u_i) - \psi_1(v_i)$. All later terms are positive, so

$$J \geq \sum_{i=1}^{13} \left(\frac{1}{u_i^2} - \frac{1}{v_i^2} \right) = \frac{2015640690251}{25971865920} > \frac{776}{10}.$$

Since $\pi > 157/50$ and $225 - J < 225$,

$$C_0 > 266 - \frac{3(225 - 776/10)}{(157/50)^2} = \frac{5451134}{24649} > \frac{88371}{400} = \frac{81}{200} C_1,$$

so $81/200 < \theta^*$. For the upper bound the thirteen demi-intervals are disjoint and ordered and ψ_1 is positive and decreasing, so the sum telescopes below its first term:

$$J < \psi_1(1/14) = 196 + \sum_{k \geq 1} \frac{1}{(k + 1/14)^2} < 196 + \frac{\pi^2}{6} < 198 < 225,$$

whence $C_0 < 266$ and $\theta^* < 532/1091 < 1/2$. So

$$\frac{81}{200} < \theta^* < \frac{1}{2}.$$

Both bounds, the definitions of C_0 , C_1 and θ^* from the trigamma series, the membership of $31/4$ and of every power of it, and the exclusion of $3/2$ are checked by the Lean kernel in the module `RationalBaseContour`, which the library root imports at the pinned source revision 92b88dc1bbe0. The lower bound on J is [one declaration](#), the rational bound on C_0 is [a second](#), the comparison $81/200 < \theta^*$ is [a third](#), the bound $\theta^* < 1/2$ is [a fourth](#), the membership of $31/4$ and of its powers are [a fifth](#) and [a sixth](#), and the exclusion of $3/2$ is [a seventh](#). Each is a finite comparison against the defined constant; none of them is any part of the analytic argument of Theorem 2.2. The sharper interval in the preceding proof is supplied by a separate exact-arithmetic certificate with 256 summands per trigamma difference. The old thousand-term decimal evaluation is not used as a proof or as a freshly reproduced receipt. The rational-bracket Lean statements and this new interval certificate have different verification histories.

2.6 Evidence, attribution and the remaining obligation

Evidence decomposition. Theorems 2.2 and 2.3 are ordinary proofs. They use Zudilin’s construction: his Lemma 7 in its polynomial reading, together with the inputs of that lemma’s own proof, namely his Lemma 3, the exponent $M(a; b)$ of (16) proved by his Lemma 4, the group stability of his Lemma 5, and the identity (9)–(11), all in [8, pp. 156–161], and his direction and constants [8, p. 162]. Proved here are the non-rationality of F as a function, the transfer of Lemma 7 to explicit integer polynomials, the exact degree computation, the degree bound on V_n , positivity, the Archimedean size estimate, and the limit $(K_n - W_n)/n^2 \rightarrow C_0$. That last limit rests on Zudilin’s Lemmas 1 and 2 [8, p. 155], whose method goes back to [7, Lemma 1, p. 466]; Section 2.3 proves them by that method with an explicit truncation estimate. The supplied public source also contains the source-applied irrationality and measure declarations, in addition to the checked finite height comparisons, the rational bracket of Section 2.5, and the degree identities of Section 2.2. The formal-source appendix distinguishes these endpoints from exact computer algebra and numerical evaluation. The supplied earlier record reports exact reconstructions of U_n, V_n for $n \leq 4$, described in Section 2.7; that reconstruction was not rerun for this revision. Fresh exact computations here are the interval enclosure above and the finite coefficient-moment and cyclotomic-residue certificates in Section 3.2. These receipts do not constitute a new Lean build. Theorem 2.2 is checked in Lean as [the rational-base region](#), with the source supply applied rather than assumed.

Attribution. Bundschuh and Väänänen proved the irrationality of this value on the region $\log b / \log a < 1/2 - 1/\pi^2 = 0.3986788163576622 \dots$ [6, Thm. 2, p. 177; hypotheses pp. 175–176], and every base of Theorem 2.2 with $b \leq 3$ is already theirs. Their printed hypothesis for $\alpha = -1$ is $\lambda < (1/2 + 1/\pi^2)^{-1}$ with $\lambda = \log h(q) / \log |q|$; at $q = a/b$ this reads $\log a / \log(a/b) < (1/2 + 1/\pi^2)^{-1}$, which is the displayed inequality. Duverney proved an independent and strictly smaller region for the same series [13, Théorème 2, p. 174]. Zudilin remarked that his generalized q -logarithm results can be given at non-integer rational bases under an assumption $\log |r| > c \log |s|$ for a computable $c > 0$, without computing a value [9, Sec. 2, p. 4]. Zudilin’s 2004 paper supplies the forms and the exponent μ under the standing hypothesis $p = 1/q \in \mathbb{Z} \setminus \{0, \pm 1\}$, and it states no rational-base result [8, Sec. 2, p. 154; Thm. 1]. The contribution here is the rational specialisation of the 2004 forms, with the denominator accounting and the limit passage carried out in full, which identifies the printed μ as an admissible c for F itself, together with the region that constant defines and its application to $31/4$.

Where the earlier criterion stops. Both regions are cut out by the same quantity. The published cutoff $1/2 - 1/\pi^2$ of [6, Thm. 2, p. 177] is the reciprocal of $\mu_{\text{BV}} = 2\pi^2/(\pi^2 - 2)$, the constant Van Assche later recovered as an integer-base irrationality-exponent bound for $F(p)$ [17, Thm. 1, p. 10], and the cutoff θ^* of Theorem 2.2 is the reciprocal of the smaller integer-base bound $\mu = C_1/C_0$ printed at [8, p. 162]. The remark following Theorem 2.5 records that reciprocal relation for every family satisfying the hypotheses of that theorem, and Zudilin’s family satisfies them with decay exponent $\sigma = C_0$ and exact degree limit $d = C_1 - C_0$. The step taken here is the homogenisation of Step 3 in

the proof of Theorem 2.2, applied to the forms that carry the smaller exponent. It substitutes one integer-base exponent bound for another, and it leaves the analytic content of each bound where its own source proves it. The bases gained are exactly the strip $s^\mu < r \leq s^{\mu_{BV}}$, which is empty for $s = 2$ and $s = 3$ and is first occupied at $s = 4$ by the single numerator 31. The kernel-checked declarations of Theorem 2.3 and of Section 2.5 decide the integer comparisons and the rational bracket that place a base inside or outside each region. They decide no step of either analytic argument, and they improve neither exponent bound.

The exact remaining obligation. Theorem 2.2 settles no base with $\log b / \log a \geq \theta^*$, and in particular settles nothing at $3/2$, whose parameter is $\log 2 / \log 3 = 0.6309297535714574\dots$, a gap of $0.2252467\dots$ above θ^* . Enlarging the region by this proof requires a smaller integer-base exponent bound carried by forms with the $\mathbb{Z}[p]$ -integrality of Lemma 7 type and with base-uniform size; a bound 2.4234 would give 0.4126 and would admit $29/4$. The source-applied irrationality and measure declarations are present in PaperR17/SourceConsumers.lean in the supplied public snapshot. The formalisation is therefore not limited to integer degree bookkeeping. This does not settle the missing approximation family at $3/2$.

Theorem 2.5 (degree-budget cap on the mechanism). *Let $(U_n, V_n) \in \mathbb{Z}[x]^2$ be a sequence such that, for constants $\sigma, \delta > 0$ and $h \geq 0$ independent of n and of the base,*

1. $\Lambda_n(x) := U_n(x)F(x) - V_n(x) \neq 0$ for every real $x > 1$;
2. $\deg U_n, \deg V_n \leq \delta n^2(1 + o(1))$;
3. the coefficient heights satisfy

$$\log \max(H(U_n), H(V_n)) \leq hn^2(1 + o(1)),$$

where $H(P)$ is the largest absolute value of a coefficient of P ;

4. the remainders satisfy

$$\log |\Lambda_n(x)| = -\sigma n^2 \log x (1 + o(1))$$

for every real $x > 1$.

Put $d_n := \max(\deg U_n, \deg V_n)$. Then $\sigma \leq \delta$, and for every fixed rational base $a/b > 1$,

$$\limsup_{n \rightarrow \infty} n^{-2} \log |b^{d_n} \Lambda_n(a/b)| \leq \delta \log b - \sigma \log(a/b).$$

Consequently the homogenised forms tend to zero whenever $\log b / \log a < \sigma / (\sigma + \delta)$, a sufficient region whose cutoff is at most $1/2$. If the actual degrees satisfy $d_n/n^2 \rightarrow d$, then $d \geq \sigma$ and the limit exists and equals $d \log b - \sigma \log(a/b)$; in that case the forms tend to zero below $\log b / \log a = \sigma / (\sigma + d)$ and their absolute values tend to infinity above it, so the exact-degree case has no decaying homogenised forms at $3/2$.

Proof. The only point not immediate from homogenisation is the cap $\sigma \leq \delta$. We apply the forms at arbitrarily large integer bases p : the upper bounds for their values and the

two-sided remainder asymptotic give an upper bound for the irrationality exponent of $F(p)$, while Dirichlet's theorem gives the lower bound 2. Letting p grow removes the base-independent height term and forces the cap.

The displayed inequality is the identity

$$n^{-2} \log |b^{d_n} \Lambda_n(a/b)| = (d_n/n^2) \log b - \sigma \log(a/b) + o(1)$$

together with hypothesis (2); the degree bound is what makes $b^{d_n} U_n(a/b)$ and $b^{d_n} V_n(a/b)$ integers.

To prove $\sigma \leq \delta$, fix an integer $p \geq 2$, write $\xi = F(p)$, $\alpha = h + \delta \log p$ and $\beta = \sigma \log p$, and set $\varepsilon_n = \Lambda_n(p)$. Hypotheses (2) and (3) give the upper bound $|U_n(p)| \leq e^{(\alpha+o(1))n^2}$, an inequality and not an equality: a degree and height bound leaves cancellation at p possible. Hypothesis (4) gives $\log |\varepsilon_n| = -\beta n^2(1 + o(1))$ on both sides, and hypothesis (1) gives $\varepsilon_n \neq 0$. Since $U_n(p)\xi - V_n(p) = \varepsilon_n$ is a nonzero real tending to 0 while $U_n(p)$ and $V_n(p)$ are integers, ξ is irrational; in particular $U_n(p) \neq 0$ for all large n , since otherwise $\varepsilon_n = -V_n(p)$ would be a nonzero integer tending to 0.

Let P/Q be rational with Q large, and let n be least with $|\varepsilon_n| < 1/(2Q)$. Minimality and the two-sided asymptotic give $n^2 \leq (1 + o(1)) \log Q/\beta$, hence

$$|\varepsilon_n| \geq Q^{-1-o(1)}, \quad |U_n(p)| \leq Q^{\alpha/\beta+o(1)}.$$

If $U_n(p)P - V_n(p)Q \neq 0$, then from

$$U_n(p)P - V_n(p)Q = Q\varepsilon_n - U_n(p)(Q\xi - P)$$

and $|Q\varepsilon_n| < 1/2$ one gets $|U_n(p)||Q\xi - P| > 1/2$, so $|\xi - P/Q| > 1/(2Q|U_n(p)|)$. If instead $P/Q = V_n(p)/U_n(p)$, then $|\xi - P/Q| = |\varepsilon_n|/|U_n(p)|$. In both cases $|\xi - P/Q| \geq Q^{-1-\alpha/\beta-o(1)}$, so the irrationality exponent satisfies $\mu(\xi) \leq 1 + \alpha/\beta$. Dirichlet's theorem gives $\mu(\xi) \geq 2$ for the irrational ξ , hence $\beta \leq \alpha$, that is $(\sigma - \delta) \log p \leq h$. The three constants do not depend on p , so letting $p \rightarrow \infty$ gives $\sigma \leq \delta$. The exact-degree statement is the same argument with $d + \varepsilon$ in place of δ for every $\varepsilon > 0$. $\square$

Corollary 2.6 (Undivided forms below the square boundary). *Under the hypotheses of Theorem 2.5, for positive integers a, b with $b < a < b^2$, the undivided forms $b^{d_n} \Lambda_n(a/b)$ do not tend to zero. No limit of d_n/n^2 is assumed.*

Proof. Write $x = a/b$ and suppose $C_n = b^{d_n} \Lambda_n(x) \rightarrow 0$. Eventual nonvanishing gives $\log |C_n| = d_n \log b + \log |\Lambda_n(x)|$ for all sufficiently large n . Also $|C_n| \leq 1$ eventually. The lower side of the remainder asymptotic therefore implies

$$d_n \log b \leq -\log |\Lambda_n(x)| = \sigma \log x n^2 + o(n^2).$$

Set $c = \sigma \log x / \log b$. Since $1 < x < b$, we have $0 < c < \sigma$, and the displayed inequality gives $d_n \leq (c + \varepsilon)n^2$ eventually for each $\varepsilon > 0$. The same family thus satisfies the cap hypotheses with degree upper rate c , its original height bound, and its original nonvanishing and remainder asymptotics at every real base greater than one. The cap yields $\sigma \leq c$, a contradiction. $\square$

The no-decay conclusion is [kernel-checked in Lean](#) under the stated all-base hypotheses.

This strengthens the failure of a sufficient-cutoff test to an exclusion of decay under the stated all-base hypotheses. It does not supply an actual-degree limit or assert divergence. If $d_n/n^2 \rightarrow d$, the separate exact-degree result gives the stronger conclusion $|b^{d_n}\Lambda_n(a/b)| \rightarrow \infty$ in the same strict region. Equality $a = b^2$ remains unclassified. The corollary concerns the undivided forms; it does not exclude base-dependent content division or other irrationality methods outside its hypotheses.

For the maximum-coefficient height convention, the passage to the ℓ^1 norm uses $\|P\|_1 \leq (\deg P + 1)H(P)$. The original quadratic degree bound makes the extra logarithm $o(n^2)$. This established height bound remains available when the degree rate is replaced by c .

Hypothesis (3) is essential and was absent from an earlier form of this statement: a bound on $\deg U_n$ alone controls $|U_n(p)|$ only through $H(U_n)$. At the boundary $\log b / \log a = \sigma / (\sigma + d)$ the normalised logarithm is zero and these hypotheses decide neither behaviour. Theorem 2.5 constrains families satisfying its hypotheses and does not exclude every possible Padé construction.

For the family of Section 2.1 the fourth hypothesis is the size estimate proved there and the second is (7). The third is proved next, so the cap applies to that family with no further assumption.

Lemma 2.7 (uniform coefficient height of the source forms). *There is a constant h with $\log \max(H(U_n), H(V_n)) \leq hn^2$ for every $n \geq 1$, where U_n and V_n are the polynomials of (6).*

Proof. Write $\|P\|$ for the sum of the absolute values of the coefficients of $P \in \mathbb{Z}[X]$, so that $H(P) \leq \|P\|$ and $\|PQ\| \leq \|P\| \|Q\|$.

By Lemma 2.1 every $v_{n,l}$ is 0 or 1, so $D_N/\Omega_n = \prod_{l \in S} \Phi_l$ over a subset $S \subseteq \{1, \dots, N\}$. Each Φ_l has Mahler measure 1, and a polynomial of degree D satisfies $\|P\| \leq 2^D M(P)$, so $\|\Phi_l\| \leq 2^{\varphi(l)}$ and

$$\left\| \frac{D_N}{\Omega_n} \right\| \leq 2^{\sum_{l \in S} \varphi(l)} \leq 2^{225n^2}.$$

For A_n , the Gaussian binomial $\begin{bmatrix} m \\ r \end{bmatrix}_X$ has nonnegative coefficients summing to $\binom{m}{r}$, so $\|c_{n,k}\| \leq \binom{k-1}{a_1-1} \binom{\beta_n - a_2 - 1}{\beta_n - k - 1} \leq 2^{2\beta_n}$ by (2), and summing the at most β_n terms of (3) gives $\|A_n\| \leq \beta_n 2^{2\beta_n}$. Hence $\|U_n\| \leq \|D_N/\Omega_n\| \|A_n\| \leq e^{O(n^2)}$ and $H(U_n) \leq e^{O(n^2)}$.

For V_n , bound it on the circle $|z| = 2$ and use Cauchy's estimate $H(V_n) \leq \max_{|z|=2} |V_n(z)|$, valid because $\deg V_n \geq 0$ and each coefficient is $|v_i| \leq 2^{-i} \max_{|z|=2} |V_n(z)|$. On that circle $|z^l - 1| \geq 2^l - 1 \geq 1$, so each of the at most $\beta_n + a_0$ inner terms of (4) has modulus at most 1; also $|c_{n,k}(z)z^{a_0k}| \leq \|c_{n,k}\| 2^{K_n}$ and $|D_N(z)/\Omega_n(z)| \leq \|D_N/\Omega_n\| 2^{225n^2} \leq e^{O(n^2)}$, while $|z^{-M_n}| \leq 1$. Multiplying the $O(n)$ bounds and using $K_n = O(n^2)$ gives $\max_{|z|=2} |V_n(z)| \leq e^{O(n^2)}$, hence $H(V_n) \leq e^{O(n^2)}$. $\square$

With Lemma 2.7, Zudilin's family satisfies all four hypotheses of Theorem 2.5, with exact degree limit $d = C_1 - C_0$ by (9), and decay exponent $\sigma = C_0$ because $\log \Lambda_n(x) = -(K_n - W_n) \log x + o(n^2)$ for each fixed real $x > 1$ by the size estimate of Section 2.4. For it $\sigma/(\sigma + d) = \theta^*$ and $(\sigma + d)/\sigma = \mu$: within this family the rational-base threshold is the

reciprocal of the integer-base irrationality-exponent bound. Lemma 2.7 is an ordinary proof and is not kernel-checked.

The region, explicitly. Among coprime a/b with $a \leq 60$ the region of Theorem 2.2 has 137 members, of which 78 are non-integral. The tightest members are $53/5$, at margin 0.000313 below θ^* , and $31/4$, at margin 0.001985; the closest miss is $52/5$ at $\theta = 0.4073243836\dots$, short by 0.001641. The bases in this region and outside the region of [6] are those in the strip $s^\mu < r \leq s^{\mu_{BV}}$, which is empty for $s = 2$ and $s = 3$, is $\{31\}$ for $s = 4$, and is $\{53, 54, 56\}$ for $s = 5$. The strip is infinite: its width $s^{\mu_{BV}} - s^\mu$ eventually exceeds s , so for every large enough denominator it contains an integer congruent to 1 modulo s . Hence $31/4$ is the member of the strip with least denominator and least numerator.

The bounded direction search. The direction $(14, 12, 14; 27)$ is Zudilin's [8, p. 162]; the scan below compares it with the other directions in a bounded box. Over the 37,533 primitive directions on Zudilin's cone with all entries at most 30, the quantity $\theta^*(\text{dir}) = C_0/C_1$ computed from his (25) and (26) has largest recorded value at $(14, 12, 14; 27)$ and its group image $(15, 12, 13; 26)$, both at $0.4056830213840605\dots$; the next value is $0.4056394327738419\dots$, at $(16, 13, 14; 28)$. Larger boxes were not scanned, and the floating-point comparisons were not interval-certified, so this is numerical evidence over that box rather than an exact optimality theorem. The enumeration ranges over integer tuples $(\alpha_0, \alpha_1, \alpha_2; \beta)$ with $1 \leq \alpha_j \leq 30$, $1 \leq \beta \leq 30$, satisfying the cone conditions of [8, Sec. 5, p. 161] and taken up to the common factor of the four entries; the objective is the ratio C_0/C_1 of the source's (25) and (26), evaluated from the exact rational breakpoints of the step-function weight and trigamma values at thirty decimal digits. The two leading tuples are exchanged by the source's permutation group, so they are one direction.

The rank-two condition should be distinguished from simultaneous approximation. Postelmans and Van Assche prove the $\mathbb{Q}$ -linear independence of $1, \zeta_q(1), \zeta_q(2)$ for $q = 1/p$ with integer $p \geq 2$ [29, Thm. 1.3, p. 3]. Their confluent multiple little q -Jacobi construction has two orthogonality conditions; its common integer normalisation and nonvanishing argument are in Section 6, especially (6.1)–(6.4). That theorem neither satisfies a universal rank-two template nor removes the integer inverse-base restriction. It is a comparison of constructions, not an application at $3/2$.

2.7 Finite receipts

Four finite computations support statements above. This paragraph records the earlier supplied receipts; those historical runs were not repeated in the present revision. The new rank-eight and interval certificates are listed separately below. The reconstruction and degree calculations use exact rational or integer arithmetic; the constant evaluation and direction scan use the numerical precision stated below. None proves a statement quantified over all indices, and none is kernel-checked.

The [public reproduction guide](#) gives the command and dependency pin for the [direction-search program](#). Run it with `--bound 30` and the output path named there; adding `--check` compares the result without replacing the [tracked receipt](#). The receipt

records mpmath 1.3.0 at thirty decimal digits. Its exact enumeration count is 37,533, but its ordering by C_0/C_1 is numerical and has no interval certificate.

The forms of Section 2.1 were reconstructed as explicit elements of $\mathbb{Z}[X]$ from cyclotomic products, without rational-function arithmetic, for $n \leq 4$. At $n = 1, 2, 3$ the reconstruction confirms $\deg A_n = K_n$ with $K_n = 587, 2264, 5032$, that Ω_n divides both $D_N A_n$ and $D_N B_n$ with zero remainder, that X^{M_n} divides both quotients, and that $W_n = \deg U_n = 333, 1315, 2944$ with $\deg V_n = W_n - 1$ and leading coefficients ± 1 . It also confirms $\sum_{l \leq N} \varphi(l) = 72, 278, 628$ and $\Sigma_n = 25, 94, 219$, hence $(K_n - W_n)/n^2 = 254, 949/4, 232$. The identity $U_n(x)F(x) - V_n(x) = x^{-M_n}(D_N/\Omega_n)(x)H_n(x)$ was then evaluated at $x = 31/4, 3$ and $7/2$ to relative accuracy below 10^{-39} at $n = 3$, with $\widehat{\Lambda}_n > 0$ at each, with $b^{W_n}U_n(a/b)$ an integer and $b^{W_n-1}U_n(a/b)$ not an integer at $31/4$ and $7/2$, and with $H_n(x)$ inside the bounds of Section 2.4.

The historical constant evaluation used ω , the thirteen intervals of Lemma 2.1 and forty-digit trigamma values. It reported

$$J = 77.94318447500909 \dots, \quad C_0 = 221.30008816500502 \dots,$$

and $C_1/C_0 = 2.46497868357497 \dots$, agreeing with the source's printed precision [8, p. 162]. The new rational interval certificate, not these reported digits, supports the numerical inequality in this revision.

The main term $K_n \log b - (K_n - W_n) \log a$ at $31/4$, computed from $\sum_{l \leq N} \varphi(l)$ and Σ_n alone, is negative from $n = 1$ and its normalisation by n^2 runs $-58.478, -10.280, -4.297, -3.863$ at $n = 1, 10, 100, 400$ against the limit $C_1 \log 4 - C_0 \log 31 = -3.718 \dots$. The elementary bound on the convergence rate is $O(\log^2(n+2)/n)$. Indeed, summing the $O(y \log(2y))$ endpoint errors over the floor blocks with $k \leq n$ gives $O(n \log^2(n+2))$; the remaining main-term tail is $O(1)$ since its summands are $O(n^2/k^3)$. The linear terms of M_n add $O(n)$. The stronger rate $O(1/n)$ does not follow from these estimates. Only the limit enters Theorem 2.2.

The supplied earlier record reports that the order and leading coefficient of V_N^* were computed exactly for $1 \leq N \leq 7$, with the following reported results:

N	1	2	3	4	5	6	7
ord	0	1	5	14	30	55	91
lc	1	6	108	4320	324000	40824000	8001504000.

3 The sharp 41/65 certificate

The numerical obstruction at $3/2$ is controlled by one small exact calculation. Writing it out is useful because the upper and lower inequalities play different roles: the upper inequality proves every strict analytic threshold below, while the lower inequality shows that 65 is the least exponent q for which $3^{41} < 2^q$. Exact failure of the rank-41 selector count one row earlier uses the additional integer comparison $2^{129} < 3^{82}$ below.

Theorem 3.1 (sharp power bracket). *One has*

$$2^{64} < 3^{41} < 2^{65}.$$

Consequently

$$\frac{41}{65} < \frac{\log 2}{\log 3}, \quad \frac{\log 3}{\log 2} < \frac{65}{41}.$$

Proof. Direct evaluation gives

$$2^{64} = 18446744073709551616,$$

$$3^{41} = 36472996377170786403,$$

$$2^{65} = 36893488147419103232.$$

The logarithmic inequalities follow by taking logarithms and dividing by the positive numbers $41 \log 3$ and $41 \log 2$, respectively. $\square$

The integer sides are checked as [the upper certificate](#) and [the sharp lower certificate](#).

The upper half is stronger than the earlier $81/200$ comparison in the direction needed at $3/2$. Combining it with the elementary bound $1/2 - 1/\pi^2 < 2/5$ gives the following exact deficits.

For $\rho, \sigma \in \mathbb{R}$, write

$$\Theta_{\text{HP}}(\rho, \sigma) = \frac{(1 + \rho^2)/2 + \sigma - 3\sigma^2/\pi^2}{(1 + \rho)^2/2 + \sigma(1 + \rho) + (1 + \rho^2)/2 + \sigma}.$$

Corollary 3.2 (height and Hankel deficits). *For every $\rho, \sigma \in \mathbb{R}$ with $0 \leq \rho$ and $1 + \rho \leq \sigma$,*

$$\frac{3}{13} < \frac{\log 2}{\log 3} - \left(\frac{1}{2} - \frac{1}{\pi^2} \right), \quad \frac{3}{13} < \frac{\log 2}{\log 3} - \Theta_{\text{HP}}(\rho, \sigma),$$

where Θ_{HP} is the rectangular exponent threshold. Moreover

$$\frac{\log 3 / \log 2 - 1}{3} < \frac{8}{41}.$$

Proof. The first inequality follows from $41/65 - 2/5 = 3/13$, Theorem 3.1, and $1/2 - 1/\pi^2 < 2/5$. The rectangular threshold is no larger than the classical margin $1/2 - 1/\pi^2$, so the second follows. The final inequality is a direct rearrangement of $\log 3 / \log 2 < 65/41$. $\square$

The uniform rectangular bound and cubic charge bound are [checked here](#) and [checked here](#).

The next statement is the source-facing charge comparison. It does not prove the degree ceilings: those come from the scalar factor and the universally forced first-order southeast-border factor in the Zudilin model. It proves that even granting those ceilings, neither extraction reaches the required $39/41$ fraction of the raw charge.

Theorem 3.3 (scalar and border charge no-go). *For every integer $N > 0$,*

$$41(N^3 - N) < 39(4N^3 - 3N^2).$$

For every integer $N \geq 2$,

$$41(2N^3 - N) < 39(4N^3 - 3N^2).$$

Hence the same strict inequalities hold with the left side replaced by $41E$ whenever, respectively, $E \leq N^3 - N$ or $E \leq 2N^3 - N$.

Proof. After subtraction, the first inequality is

$$N(115N^2 - 117N + 41) > 0,$$

which holds for $N > 0$. The second becomes

$$N(74N^2 - 117N + 41) > 0.$$

For $N \geq 2$ the quadratic factor is positive and increasing. The assertions for E follow by monotonicity. $\square$

The application-facing downward-closed forms are [the scalar no-go](#) and [the scalar-plus-border no-go](#).

The formal source checks Theorem 3.1, Corollary 3.2, and Theorem 3.3 as exact Lean propositions. Their role is exclusion: scalar content and the forced first-order border do not supply enough charge. Higher residual valuations, determinant cancellation, a different integral model, and irrationality of $F(3/2)$ remain open.

3.1 The sharp q -order of the normalised Hankel determinant

The charge comparison of Corollary 3.2 concerns the stated denominator model. The next theorem determines a different quantity, the formal q -order of Zudilin's normalised Hankel determinant at $x = z = 1$; converting a q -order into a denominator saving at a rational specialisation needs a separate integral normalisation and divisibility argument. Zudilin builds the determinant by combining the Padé-type approximations of Bundschuh and Zudilin [11] with Bézivin's method as developed in [12]; see [9, Sec. 2, p. 3]. His passage from the q -order to the size of the determinant borrows the proofs of [12, Lemma 2, p. 12, and Prop. 3, p. 13], as he notes in [9, Sec. 4, p. 7].

Theorem 3.4 (sharp all-rank q -order and leading coefficient). *For every rank N , the normalised Hankel determinant V_N^* of [9, Sec. 4, (6), p. 6], evaluated at $x = z = 1$, has*

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6}, \quad \text{leading coefficient} \quad \frac{(N!)^2(N+1)!}{2^N}.$$

The source proves the inequality $\text{ord}_q V_N^* \geq N(N-1)(2N-1)/6$ and uses it as an upper bound on $|V_N^*|$ [9, Sec. 4, p. 7]. Equality rules out any further initial power of q in this formal determinant at $x = z = 1$. The size at a fixed real q is a separate statement; the positive-measure estimate below proves that $V_N^*(q) > 0$ and that $\log V_N^*(q)$ differs from the logarithm of its leading term by $O_q(N)$.

The row identity. Work over $A = \mathbb{Z}[[q]]$. Let $H(X) = 1 + \sum_{s \geq 1} a_s(q)X^s$ and put

$$W_m(t) = q^{(m+1)t} \prod_{r=1}^m H(q^r), \quad D_j = \prod_{r=0}^{j-1} (I - q^r \mathcal{N}), \quad E(m, j) = mj - \frac{j(j-1)}{2},$$

where $\mathcal{N}$ is the backward shift $(\mathcal{N}f)_m = f_{m-1}$ in the index m , so D_j is the source's backward-difference operator of [9, Sec. 4, (7), p. 6]. Write $\bar{H} = H \bmod q$ and $h_r = [X^r]\bar{H}(X)^{-1}$, with $h_r = 0$ for $r < 0$ and $h_0 = 1$.

Lemma 3.5 (all-depth initial monomial). For $m \geq j \geq 0$ one has $D_j W_m(t) \in q^{E(m,j)} A$ and

$$[q^{E(m,j)}] D_j W_m(t) = (-1)^j h_{j-t}.$$

Proof. Define the following transition coefficients on states $(e_u)_{u \geq 0}$. Powers Φ^j mean the finite-depth path sums, not endomorphisms of the algebraic direct sum: for fixed initial state t , endpoint u and depth j , all visited states are at most $\max(t, u + j)$, so each such coefficient is a finite sum. Subsequent sums over endpoints are interpreted coefficientwise as justified below. The transition coefficients are

$$\Phi e_0 = \sum_{s \geq 1} a_s e_{s-1}, \quad \Phi e_u = (q^u - 1)e_{u-1} + q^u \sum_{s \geq 1} a_s e_{u+s-1} \quad (u > 0).$$

A direct expansion gives

$$W_m(u) - W_{m-1}(u) = q^m \sum_v (\Phi e_u)_v W_{m-1}(v),$$

since both sides equal

$$q^{mu} \prod_{r < m} H(q^r) (q^u H(q^m) - 1).$$

Using $r + E(m-1, r) = E(m, r)$, induction on j turns this into

$$D_j W_m(t) = q^{E(m,j)} \sum_u (\Phi^j e_t)_u W_{m-j}(u). \quad (12)$$

The sum is locally finite: one transition lowers a state by at most one, so a depth- j path ending at u never visits a state above $\max(t, u + j)$, and $\text{ord } W_{m-j}(u) = (m - j + 1)u$, so only finitely many endpoints contribute to any fixed power of q .

Modulo q the operator simplifies: $\bar{\Phi} e_u = -e_{u-1}$ for $u > 0$, and $\bar{\Phi} e_0 = \sum_{s \geq 1} a_s(0) e_{s-1}$. Put $c_{j,t} = (\bar{\Phi}^j e_t)_0$. Then $c_{j+1,t} = -c_{j,t-1}$ for $t > 0$ and $c_{j+1,0} = \sum_{s=1}^{j+1} a_s(0) c_{j,s-1}$, the sum terminating because a state above j cannot reach 0 in j steps. Now $c_{0,t} = \delta_{t,0} = h_{-t}$, and if $c_{j,t} = (-1)^j h_{j-t}$ for all t then $c_{j+1,t} = (-1)^{j+1} h_{j+1-t}$ for $t > 0$ at once, while for $t = 0$ the identity $\bar{H} \bar{H}^{-1} = 1$ gives $h_{j+1} = -\sum_{s \geq 1} a_s(0) h_{j+1-s}$ and hence $c_{j+1,0} = (-1)^j \sum_{s \geq 1} a_s(0) h_{j+1-s} = (-1)^{j+1} h_{j+1}$. So $c_{j,t} = (-1)^j h_{j-t}$ for all j, t . Since $\text{ord } W_{m-j}(u) = (m - j + 1)u$ and $m \geq j$, in (12) only the state $u = 0$ contributes at degree $E(m, j)$, which gives both assertions. $\square$

Apply this to the normalised source tail

$$T_{m,t} = q^{(m+1)t} (q; q)_m^3 \frac{(q^{t+1}; q)_m}{(q^{m+t+1}; q)_{m+1}}.$$

With $C_t = (1 - q^{t+1})^{-1}$ and

$$H_t(X) = \frac{(1 - X)^3 (1 - q^t X)^2}{(1 - q^t X^2)(1 - q^{t+1} X^2)}$$

one has $T_{m,t} = C_t W_m^{H_t}(t)$, by the telescoping identity

$$\prod_{r \leq m} H_t(q^r) = \frac{(q; q)_m^3 (1 - q^{t+1})(q^{t+1}; q)_m}{(q^{m+t+1}; q)_{m+1}}.$$

The two denominator products contribute the consecutive factors $1 - q^{t+2}, \dots, 1 - q^{t+2m+1}$. Reducing modulo q gives $\bar{H}_t = (1 - X)^3$ for $t > 0$, so $h_r^{(t)} = \binom{r+2}{2}$, and $\bar{H}_0 = (1 - X)^4/(1 + X)$, so $h_r^{(0)} = (r+1)(r+2)(2r+3)/6$. Both have $C_t(0) = 1$. The rows are $v_m = \sum_{t \geq 0} T_{m,t}$, and interchanging that sum with the finite difference D_j is legitimate because $\text{ord } T_{m,t} = (m+1)t$. Terms with $t > j$ contribute $h_{j-t}^{(t)} = 0$, so Lemma 3.5 at $m = j + \ell$, where $E(j + \ell, j) = j(j+1)/2 + j\ell$, gives

$$D_j v_{j+\ell} = (-1)^j \frac{(j+1)^2(j+2)}{2} q^{j(j+1)/2+j\ell} + O(q^{j(j+1)/2+j\ell+1}) \quad (j, \ell \geq 0), \quad (13)$$

the coefficient because

$$h_j^{(0)} + \sum_{t=1}^j h_{j-t}^{(t)} = \frac{(j+1)(j+2)(2j+3)}{6} + \binom{j+2}{3} = \frac{(j+1)^2(j+2)}{2}.$$

Proof of Theorem 3.4. The operators D_j act by lower unitriangular row operations, so they leave $\det(v_{i+j})_{0 \leq i, j < N}$ unchanged. By (13) the entry in row j and column ℓ has order $e(j, \ell) = j(j+1)/2 + j\ell$. In the Leibniz expansion the weight of a permutation ς is $\sum_j (j(j+1)/2 + j\varsigma(j))$, and by the rearrangement inequality $\sum_j j\varsigma(j)$ is uniquely minimised by the reversal $\varsigma(j) = N - 1 - j$, the values j being distinct. The minimum weight is $\sum_{j < N} j^2 = N(N-1)(2N-1)/6$, so exactly one Leibniz term attains it and no cancellation is possible there. The sign of the reversal is $(-1)^{N(N-1)/2}$, which cancels $\prod_{j < N} (-1)^j$, and the surviving coefficient is

$$\prod_{j=0}^{N-1} \frac{(j+1)^2(j+2)}{2} = \frac{(N!)^2(N+1)!}{2^N}.$$

□

Formal order and analytic size are different questions. Theorem 3.4 fixes the first nonzero power of q and its coefficient at each fixed rank. It does not control the value at a fixed rational q as the rank grows. For the logic, the integer polynomials $f_N(q) = C_N q^{B_N} (1 - q)^{N^3}$, with $B_N = \sum_{j < N} j^2$ and $C_N = (N!)^2(N+1)!/2^N$, have exactly the order B_N and exactly the leading coefficient C_N , while $f_N(2/3) = C_N (2/3)^{B_N} 3^{-N^3}$ carries a further cubic exponential factor that neither datum sees. The following separate positive-measure argument supplies the fixed-base estimate for V_N^* ; it is not inferred from the formal order.

A separate positive-measure estimate. For fixed $0 < q < 1$ write $P = (q; q)_\infty$, $Q = (\sqrt{q}; q)_\infty$, $T = (-1; q)_\infty^2$, and

$$G_q(w) = \frac{1}{(w; q)_\infty^3} \sum_{t \geq 0} \frac{w^t}{(q; q)_t} \frac{(q^t w^2; q)_\infty}{(q^t w; q)_\infty^2}, \quad \gamma_k = [w^k] G_q(w), \quad c_k = \frac{(k+1)^2(k+2)}{2}.$$

All products converge normally on $|w| \leq r < 1$ and the t -sum is bounded there by a constant times $\sum_t r^t$. The finite-to-infinite product identities give $v_m^* = P^4 G_q(q^{m+1})$. The q -binomial theorem [27, Eq. 17.2.37]

$$\frac{(Aw; q)_\infty}{(w; q)_\infty} = \sum_{j \geq 0} \frac{(A; q)_j}{(q; q)_j} w^j \quad (0 \leq A \leq 1)$$

follows by comparing coefficients in $(1 - w)R(w) = (1 - Aw)R(qw)$ with $R(0) = 1$; the series converges for $|w| < 1$ since its coefficients are at most P^{-1} . They are nonnegative, and at least $(A; q)_\infty$ if $A < 1$. For $A = 1$ the series equals 1. Factor the numerator of the t th term using

$$(q^t w^2; q)_\infty = (q^{t/2} w; q)_\infty (-q^{t/2} w; q)_\infty (q^{(t+1)/2} w; q)_\infty (-q^{(t+1)/2} w; q)_\infty.$$

Pair the two positive-argument factors with two copies of $(w; q)_\infty^{-1}$. All remaining factors have nonnegative coefficients. The $t = 0$ term bounds γ_k below by $Q \binom{k+3}{3}$. For upper bounds, a nonnegative-coefficient regular factor R with $R(1) \leq C$, multiplying a series with nondecreasing coefficients d_k , contributes at most $C d_k$ by convolution. This rule bounds the $t = 0$ term by $TP^{-4} \binom{k+3}{3}$ and the sum of the $t \geq 1$ terms by $TP^{-6} \binom{k+2}{3}$. Consequently

$$\frac{Q}{3} c_k \leq \gamma_k \leq T(P^{-4} + P^{-6}) c_k.$$

Thus $\sum_{k \geq 0} P^4 \gamma_k q^k \delta_{q^k}$ is a finite positive measure with infinitely many distinct support points and moments v_m^* . A nonzero polynomial of degree less than N cannot vanish at all of its first N atoms; the associated Gram matrix is positive definite. Finite Cauchy–Binet followed by convergence of every matrix entry and monotone convergence of the nonnegative tuple sums gives Heine’s expansion of a Hankel determinant of moments, whose integral form is reproved in [10, Sec. 2, (2)–(5), pp. 2–3]:

$$V_N^* = \sum_{k_0 < \dots < k_{N-1}} \prod_i (P^4 \gamma_{k_i} q^{k_i}) \prod_{i < j} (q^{k_i} - q^{k_j})^2.$$

Retaining the tuple $k_i = i$ and using $\prod_{d=1}^{N-1} (1 - q^d)^{2(N-d)} \geq P^{2N}$ gives the lower bound below. For the upper bound put $k_i = i + \lambda_i$, use $c_{i+\lambda}/c_i \leq (\lambda + 1)^3$, discard residual Vandermonde factors bounded by 1, and drop the ordering restriction on the nonnegative λ_i . With $S(q) = (1 + 4q + q^2)/(1 - q)^4$ this gives

$$(P^6 Q/3)^N C_N q^{B_N} \leq V_N^*(q) \leq [P^4 T(P^{-4} + P^{-6}) S(q)]^N C_N q^{B_N}.$$

Both constants are positive and finite, so $V_N^*(q) > 0$ and $\log(V_N^*(q)/(C_N q^{B_N})) = O_q(N)$, including $N = 0$ under the empty determinant convention. The fixed- q constants give an error linear in N after taking logarithms; they are not uniform as $q \uparrow 1$. The measure depends on q but not on the moment index m or the Hankel rank N . This is a separate analytic proof; it supplies no new denominator factor for the 2004 polynomial forms. The generating-function identification and infinite determinant expansion in this argument are not yet formalised in Lean.

Evidence decomposition. The positive-measure proof above is ordinary mathematics and is not a Lean claim. Separately, the formal-order proof has the supplied Lean check of [its all-depth associated-graded recurrence](#) and source-applied all-rank conclusions for the formal order and initial coefficient. These are not conclusions about the real measure. The order and the leading coefficient of V_N^* are [the sharp Hankel order and coefficient](#), and the two closed forms $6 \text{ ord} = N(N-1)(2N-1)$ and $2^N \text{ lc} = (N!)^2(N+1)!$ are [the closed order](#) and [the closed leading coefficient](#). The Lean proof composes the initial monomial of every transformed row, which for row 1 has order exactly $l+1$ and coefficient exactly -6 in every column. Independently of the proof, the order and the leading coefficient were computed exactly for $1 \leq N \leq 7$, giving orders 0, 1, 5, 14, 30, 55, 91 and leading coefficients 1, 6, 108, 4320, 324000, 40824000, 8001504000, each matching the closed forms.

Attribution and remaining obligation. The antecedent is the inequality of [9, Sec. 4, p. 7]; the equality and the leading coefficient are proved here. Lemma 3.5 and (13) give the initial monomial of every row, and Lean states that identity for every row as [the all-row initial monomial](#). Nothing in this subsection decides the arithmetic nature of $F(3/2)$.

3.2 Coefficient moments and cyclotomic content

The positive measure just constructed belongs to the remainders v_m^* . It is not a measure for their coefficients in $F(p)$. To make the latter question precise, fix $p > 1$, use Gaussian binomial polynomials, and put

$$R_m(p) = (-1)^m P_m(p^{m+1} | p^{-1}) = \sum_{k=0}^m (-1)^{m+k} p^{k(k+1)/2} \begin{bmatrix} m \\ k \end{bmatrix}_p \begin{bmatrix} m+k \\ k \end{bmatrix}_p, \quad s_m(p) = ([m]_p!)^3 R_m(p),$$

where $[m]_p! = \prod_{j=1}^m (1 + p + \dots + p^{j-1})$. The polynomial normalisation of [9, Sec. 3, p. 5], at $x = z = 1$, gives

$$\alpha_m = p[p(p-1)^3]^m s_m(p), \quad \beta_m = \left[\alpha_m \sum_{j \geq 1} \tau(j) p^{-j} \right]_+ - 1, \quad v_m^* = \alpha_m F(p) - \beta_m.$$

The brackets mean the polynomial part at infinity. Thus $\alpha_0 = p$ and $\beta_0 = 0$. This fixes the normalisation before any content or positivity test.

A positive expansion is not a moment representation. Substituting $x = p^{m+1}$ into Van Assche's alternative expansion [17, (16), p. 4] gives the identity

$$R_m(p) = \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_p \begin{bmatrix} m+k \\ k \end{bmatrix}_p p^{(m-k)(m-k+1)/2} \prod_{j=m-k+1}^m (p^j - 1).$$

Indeed $(qx; q)_k = (-1)^k \prod_{j=m-k+1}^m (p^j - 1)$, so the two signs cancel. This proves $R_m(p) > 0$ for $p > 1$, and nonnegative coefficients in the variable $t = p - 1$; it does not prove Hankel positivity. In fact,

$$\det(R_{i+j}(p))_{i,j < 3} = -36(p-1)^2 + O((p-1)^3) \quad (p \downarrow 1).$$

The cubed factorial is therefore not an inessential positive factor: entrywise multiplication by a positive sequence need not repair a failed moment condition. At $p = 1$, $s_m(1) = (m!)^3$ is the moment sequence of a product of three independent unit exponential variables, since their m th moments multiply. This is only the polynomial endpoint; $F(1)$ diverges. Berg's Theorem 5.1 states that $(m!)^c$ is Stieltjes indeterminate for $c > 2$ [34]. Thus a measure exists at the endpoint but is not unique. Neither uniqueness nor a canonical deformation is a premise of the question at $p > 1$.

Finite certificates and an all-rank degree check. Put $D_{N,h}(p) = \det(s_{i+j+h}(p))_{0 \leq i,j < N}$, with $D_{0,h} = 1$. Exact polynomial calculations give strictly positive coefficients in $t = p - 1$ for $D_{N,h}(1 + t)$ when $h = 0, 1$ and $1 \leq N \leq 8$. Thus sixteen determinant polynomials are positive for every real $p \geq 1$. Their degrees, in increasing rank, are

$$\begin{array}{l|cccccccc} h = 0 & 0 & 10 & 54 & 156 & 340 & 630 & 1050 & 1624 \\ h = 1 & 2 & 26 & 96 & 236 & 470 & 822 & 1316 & 1976. \end{array}$$

The complete coefficient lists, polynomial definitions and an exact reproduction script accompany the revision. This is finite symbolic verification, not sampling of bases and not a Lean theorem. It is also not a test of every Hankel minor or of arbitrary rank.

The degrees admit a separate all-rank proof. The Gaussian binomial $\begin{bmatrix} u \\ v \end{bmatrix}_p$ is monic of degree $v(u - v)$. In the signed sum for R_m , the degree increases by $2m - k$ from the k th to the $(k + 1)$ st term. The unique maximal term is $k = m$, with positive leading coefficient. Hence R_m is monic of degree $(3m^2 + m)/2$, and s_m is monic of degree $3m^2 - m$. In the determinant, the only permutation-dependent part of the degree is $6 \sum_i i\sigma(i)$; strict rearrangement makes the identity its unique maximum. Therefore

$$\deg D_{N,h} = \sum_{i=0}^{N-1} (3(2i + h)^2 - (2i + h)), \quad \text{lc } D_{N,h} = 1.$$

More generally the same argument works for a fixed minor with distinct increasing row and column indices. Each such minor is positive for all sufficiently large p , with a threshold that may depend on the minor. At $p = 1$, both leading Hankel families are positive definite by the infinite-support product measure, so each fixed rank is also positive in some neighbourhood of 1. Neither argument provides a neighbourhood or large-base threshold uniform over every rank.

The classical Stieltjes criterion requires positive semidefiniteness of both Hankel families at all ranks [31, Lemma 2.1, p. 4]. Coefficientwise total positivity would be stronger than the sixteen certificates above. Positive production matrices and path constructions can supply such a mechanism in other families [36, Sec. 2][37, Thms. 9.7–9.8]; no such matrix for this moving-degree sequence has been established here.

A finite spectral consequence. Write $A_N = (\alpha_{i+j})$ and $B_N = (\beta_{i+j})$. Let $c = p(p - 1)^3$ and $D = \text{diag}(1, c, \dots, c^{N-1})$. Then $A_N = pD(s_{i+j})D$. For $p > 1$ this is an invertible positive diagonal congruence. The remainder representation already proves $F(p)A_N - B_N = (v_{i+j}^*) > 0$ at every rank.

Proposition 3.6 (finite coefficient positivity and pencil roots). *For every real $p > 1$ and $1 \leq N \leq 8$, A_N is positive definite and all roots of $\det(YA_N - B_N)$ are real and strictly less than $F(p)$. The roots at consecutive ranks $N, N+1 \leq 8$ interlace non-strictly.*

Proof. The leading principal minors of $(s_{i+j})_{i,j < N}$ are exactly $D_{k,0}(p)$ for $1 \leq k \leq N$; their certified positivity and Sylvester's criterion give positive definiteness. The diagonal congruence gives $A_N > 0$. The real symmetric matrix $A_N^{-1/2}B_N A_N^{-1/2}$ has these pencil roots as eigenvalues, and $F(p)I - A_N^{-1/2}B_N A_N^{-1/2} > 0$ puts them strictly below $F(p)$. The minimum–maximum principle for $x^T B_N x / (x^T A_N x)$ on nested coordinate subspaces gives non-strict interlacing. $\square$

The proposition uses the finite certificates, not an all-rank coefficient measure. Such a measure of infinite support would extend the same proof to all ranks. The largest root is a nondecreasing lower bound over the certified ranks, but convergence to $F(p)$, simple roots, strict interlacing and denominator control do not follow. At $p = 1$ the congruence degenerates and F diverges, so the proposition excludes that endpoint. This pencil is not automatically the Jacobi matrix of the coefficient moments.

A more concrete coefficient-moment question. Write $D_n = D_{n,0}$, $E_n = D_{n,1}$, with $D_0 = E_0 = 1$. The classical formal Stieltjes continued fraction has determinant ratios

$$\lambda_{2n-1} = \frac{E_n D_{n-1}}{D_n E_{n-1}}, \quad \lambda_{2n} = \frac{D_{n+1} E_{n-1}}{E_n D_n} \quad (n \geq 1).$$

The classical correspondence is recalled in [35, pp. 1–2]. Our certificates imply $\lambda_1, \dots, \lambda_{15} > 0$ for every $p \geq 1$. Seeking a positive formula or production mechanism for these ratios at every index is more specific than asking whether a measure might exist. The generating function in this correspondence is formal: at $p = 1$ its coefficients $(m!)^3$ already give radius of convergence zero.

The same two positive 8×8 moment matrices give a finite positive quadrature realisation for $s_0, \dots, s_{15}$, by the classical orthogonal-polynomial/Jacobi construction; the computation of nodes and weights from that matrix is described by Golub and Welsch [39, Sec. 2, (2.6); Sec. 4]. This is a truncated moment statement, not a measure representing all s_m . The Jacobi construction here must not be identified with the different coefficient pencil (A_N, B_N) . For rational modifications of an already known moment functional, see Krattenthaler [30, Thm. 1; Sec. 6]; such a modification producing the entire sequence s_m has not been identified.

What would count as extra cyclotomic divisibility. Independently of the moment question, define

$$\mathcal{H}_N(Y; p) = \det(\alpha_{i+j}Y - \beta_{i+j})_{i,j < N}, \quad t_{m,d} = \min(v_{\Phi_d}(\alpha_m), v_{\Phi_d}(\beta_m)), \quad e_{N,d} = \min_{\sigma \in S_N} \sum_{i=0}^{N-1} t_{i+\sigma(i),d},$$

with $v_{\Phi_d}(0) = \infty$. The valuation of a polynomial in Y means the minimum valuation of its coefficients. Each determinant term is divisible by $\Phi_d^{e_{N,d}}$. Set

$$\Gamma_{N,d}(Y) = \sum_{\substack{\sigma \in S_N \\ \sum_i t_{i+\sigma(i),d} = e_{N,d}}} \text{sgn}(\sigma) \prod_i \overline{\Phi_d^{-t_{i+\sigma(i),d}}(\alpha_{i+\sigma(i)}Y - \beta_{i+\sigma(i)})},$$

where the bar is reduction in $(\mathbb{Q}[p]/(\Phi_d))[Y]$. Then $v_{\Phi_d} \mathcal{H}_N = e_{N,d}$ exactly when $\Gamma_{N,d} \neq 0$. This isolates the cancellation that a genuine additional factor would require. A useful determinant form of this test is available whenever integers u_i, v_j satisfy $u_i + v_j \leq t_{i+j,d}$ and $\sum_i u_i + \sum_j v_j = e_{N,d}$. Divide row i by $\Phi_d^{u_i}$ and column j by $\Phi_d^{v_j}$ in the local fraction field, then reduce modulo Φ_d . Only entries with $u_i + v_j = t_{i+j,d}$ survive. The determinant of this reduced matrix is $\Gamma_{N,d}$, since every non-minimal permutation vanishes on reduction. Thus a root-of-unity operator annihilating the surviving matrix would explain excess valuation; its hypotheses must be checked for this specific pencil. This assertion is conditional on the displayed potentials and does not assume that the matrix has rank one.

The monic contents in $\mathbb{Q}[p]$, computed exactly, are

N	$\text{cont}_Y \mathcal{H}_N$
1	p
2	$p^5(p-1)^4$
3	$p^{14}(p-1)^{15}(p+1)^4$
4	$p^{30}(p-1)^{32}(p+1)^8(p^2+p+1)^4$
5	$p^{55}(p-1)^{55}(p+1)^{19}(p^2+1)^4(p^2+p+1)^8$.

They attain the assignment bound for every d at these five ranks: $d \leq 8$ was tested directly, and the full displayed factorisations exclude other cyclotomic factors. The power of p is not cyclotomic content. Since $\deg_Y \mathcal{H}_N \leq N$, its coefficient gcd over $\mathbb{Q}[p]$ equals the gcd of its values at $Y = 0, \dots, N$, by invertibility of the rational Vandermonde matrix. This is the computation used here; it does not certify specialised integer contents or all-rank behaviour.

A separate minimum-cost subset recurrence now gives nonzero residue witnesses for every pair

$$1 \leq N \leq 8, \quad 1 \leq d \leq \max(8, 2N - 2).$$

There are 76 such pairs. For each, the supplied record gives an integer $Y_0 \in \{0, \dots, N\}$ and a nonzero residue $\Gamma_{N,d}(Y_0) \bmod \Phi_d$, which certifies equality with the assignment bound. At ranks six to eight this is a finite cyclotomic-index window, not a complete content factorisation. The complete contents above remain certified only through rank five.

These computations distinguish systematic common factors from cancellation beyond entry valuations, a distinction also important in fraction-free matrix decompositions [38, Sec. 2; Thm. 6]. In particular the gcd argument at the points $0, \dots, N$ is over $\mathbb{Q}[p]$: the rational Vandermonde inverse is not generally integral, so it must not be used to assert the corresponding coefficient gcd in $\mathbb{Z}[p]$.

Proposition 4 of Krattenthaler–Rochev–Väänänen–Zudilin [12, pp. 14–15; (4.10), p. 17] obtains cyclotomic factors through root-of-unity annihilation for a different first-order tail recurrence. No corresponding recurrence has been established for this moving-diagonal pencil. The finite witnesses show no excess in their stated range, not the impossibility of excess at larger ranks or other cyclotomic indices. Whether

minimum-valuation residues ever cancel systematically, and whether s_m is a Stieltjes sequence, remain separate questions. The remainder measure answers neither.

4 Integer scalar content is neutral

An irrationality argument by linear forms replaces the clearing scheme by explicit rational approximation: one constructs nonzero linear forms in 1 and $F(\beta)$ with integer coefficients whose absolute values tend to zero. Bounds on coefficient height help construct those forms or bound an irrationality exponent, but height times error need not tend to zero for the one-form irrationality criterion. For an integer coefficient pair (U, V) and a real target S , put

$$L_S(U, V) = US - V, \quad \Delta((U_n, V_n), (U_m, V_m)) = U_n V_m - U_m V_n.$$

We call $L_S(U, V)$ the *error* of the row at S ; a good approximation is one that makes it small. The second expression is the exterior determinant of the two rows, and it eliminates S exactly:

$$\Delta = U_m L_S(U_n, V_n) - U_n L_S(U_m, V_m).$$

This identity is what makes the determinant useful. Since Δ is an integer, if it does not vanish then

$$1 \leq |\Delta| \leq |U_m| |L_S(U_n, V_n)| + |U_n| |L_S(U_m, V_m)|,$$

and the two errors cannot both be smaller than $1/(|U_n| + |U_m|)$. The determinant therefore carries two competing quantities at once, an integer that divides it and its own absolute value, and the theorem below is about how a rescaling moves them.

Theorem 4.1 (integer-scalar-content no-go). *Let S be real, let (U_n, V_n) and (U_m, V_m) be pairs of integers, and let c_n, c_m be integers. Then*

$$L_S(c_n U_n, c_n V_n) = c_n L_S(U_n, V_n),$$

$$\Delta(c_n(U_n, V_n), c_m(U_m, V_m)) = c_n c_m \Delta((U_n, V_n), (U_m, V_m)),$$

and consequently

$$|\Delta(c_n(U_n, V_n), c_m(U_m, V_m))| = |c_n| |c_m| |\Delta((U_n, V_n), (U_m, V_m))|.$$

In particular $c_n c_m$ divides the scaled determinant. Hence a local divisor supplied only by the two integer scalar factors is paid for by exactly the same factor in the Archimedean determinant height.

Proof. All three identities are routine expansions in $\mathbb{Z}$ or $\mathbb{R}$; the divisibility statement uses the primitive determinant as its witness. $\square$

Informally, Theorem 4.1 says that multiplying specialised integer rows by scalar factors moves the local gain and the Archimedean cost by exactly the same amount. Within an argument whose only extra divisor is integer scalar content, primitive normalisation therefore loses no net gain. This says nothing about polynomial factors before specialisation, cross-row common factors, determinant-specific arithmetic, or additive combinations.

Example 4.2. Take $(U_n, V_n) = (1, 2)$ and $(U_m, V_m) = (3, 5)$, so that $\Delta = 1 \cdot 5 - 3 \cdot 2 = -1$. Multiplying the first row by $c_n = 6$ and the second by $c_m = 10$ gives the rows $(6, 12)$ and $(30, 50)$, whose determinant is $6 \cdot 50 - 30 \cdot 12 = -60$. That determinant is now divisible by 60, which looks like a local gain of 60; and its absolute value has risen from 1 to 60, which is a cost of exactly the same size.

Lean checks the error identity in [error scaling](#), the determinant identity in [content factorisation](#), the exact absolute-height identity in [absolute determinant scaling](#), and the divisor statement in [content-product divisibility](#). The elimination identity is the checked [exterior determinant identity](#).

The key point is that the two scalings are the same scaling. Each identity on its own is a one-line expansion, and the interest of the theorem is not in any one of them. Taken together they say that the factor $c_n c_m$ which a rescaling introduces into the determinant reappears undiminished, as $|c_n| |c_m|$, in the absolute value of that determinant. The third identity is displayed with absolute values for exactly that reason: it is what makes the statement one about the Archimedean height and not about divisibility alone. Whatever c_n and c_m are, a rescaling therefore leaves the balance between the local divisor and that height where it was.

The theorem does not construct primitive Padé rows, estimate their remainders, or prove that their exterior determinant is nonzero. It removes one source of apparent gain: multiplying a useful row by a large common integer cannot improve the local-to-Archimedean balance. Within a construction whose proposed gain comes only from those integer scalars, the rows may be primitive-normalised without a net loss. This does not say that every candidate family must use such a normalisation or that every proof needs separate 2-adic and 3-adic gain. The theorem quantifies over arbitrary integer pairs, so it applies whenever a construction has reached that specialised-row stage.

5 Endpoint residues at $(3, 2)$ and the four-jet kernel

Zudilin's treatment of q -harmonic series [8, Secs. 3–5] builds linear forms of the shape used in Section 4 out of Heine's basic transform. The generic endpoint and jet lemmas below concern arbitrary integral coefficient pairs of this shape; they do not construct or instantiate Zudilin's actual polynomial family. Each lemma uses no property beyond integrality and quantifies over arbitrary elements of $\mathbb{Z}[X]$, and the tools are elementary congruences, the pigeonhole principle and Bézout's identity. The exception is Theorem 5.12 at the end of the section, whose subject is the scalar parameters of Zudilin's cone. It says nothing about the coefficient polynomials.

Substituting $X = 3/2$ into an integer polynomial produces a rational number, and multiplying by a power of 2 clears its denominator. The following evaluation records that cleared numerator, so that all the arithmetic below stays inside $\mathbb{Z}$. For $P(X) = \sum_i p_i X^i \in \mathbb{Z}[X]$ and a declared width $W \geq 0$, put

$$H_W(P) = \sum_{i=0}^W p_i 3^i 2^{W-i}.$$

This is the [homogeneous endpoint evaluation](#). It is homogeneous in the sense that X^i is replaced by $3^i 2^{W-i}$, so the numerator and the denominator of the base are carried symmetrically. When $W \geq \deg P$ it is exactly the cleared numerator, since

$$\sum_{i=0}^W p_i 3^i 2^{W-i} = 2^W \sum_{i=0}^W p_i \left(\frac{3}{2}\right)^i = 2^W P\left(\frac{3}{2}\right).$$

Theorem 5.1 (endpoint residues). *Let $P = \sum_i p_i X^i \in \mathbb{Z}[X]$ and let $W \geq 0$. Then*

$$H_W(P) \equiv p_0 2^W \pmod{3}, \quad H_W(P) \equiv p_W 3^W \pmod{2}.$$

Consequently a unit constant coefficient prevents divisibility by 3, and a unit coefficient at the declared width W prevents divisibility by 2.

Proof. Modulo 3, every summand with $i > 0$ vanishes; modulo 2, every summand with $i < W$ vanishes. The remaining powers are units in the corresponding residue fields. $\square$

Since 2^W is invertible modulo 3 and 3^W is invertible modulo 2, the two congruences say more than the stated consequence: divisibility of $H_W(P)$ by 3 is decided by p_0 alone, and divisibility by 2 by p_W alone. The rest of the coefficient vector is invisible to both primes. The coefficient p_W is the top coefficient of P exactly when $\deg P = W$, and is zero when $\deg P < W$; the identity $H_W(P) = 2^W P(3/2)$ likewise holds only for $\deg P \leq W$.

Example 5.2. Take $W = 2$. The three polynomials below differ only at an endpoint.

P	$H_2(P)$	$3 \mid H_2(P)$	$2 \mid H_2(P)$
$X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 13$	no	no
$X^2 + 3$	$3 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 21$	yes	no
$2X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 2 \cdot 9 = 22$	no	yes

The first has both endpoints equal to 1 and its evaluation, 13, is divisible by neither prime; as a check, $2^2((3/2)^2 + 1) = 13$. The second and third show that each hypothesis is used: spoiling the constant endpoint admits 3, and spoiling the top endpoint admits 2.

The congruences are the [bottom-endpoint identity](#) and [top-endpoint identity](#); the unit consequences are the [constant-endpoint obstruction](#) and [top-endpoint obstruction](#).

The proof is two lines, but the shape of the statement is not incidental. The difficulty lies in the fact that the specialisation sees a different endpoint at each of the two primes: modulo 3 only the constant coefficient survives, and modulo 2 only the top one does. The two exclusions are therefore conditions at opposite ends of the coefficient vector, and the statement below imposes one at each end, on the two entries of a single coefficient pair.

Proposition 5.3 (common divisor). *Let $U, V \in \mathbb{Z}[X]$ and let $W \geq 0$. If U has unit top endpoint, V has unit constant endpoint, and an integer c divides both $H_W(U)$ and $H_W(V)$, then*

$$2 \nmid c \quad \text{and} \quad 3 \nmid c.$$

This is the checked [common-divisor exclusion](#). The proposition does not say that the two evaluations are coprime; it says that whatever they share misses both of the primes that matter at $3/2$.

Example 5.4. Take $W = 2$, $U = X^2 + 3$ and $V = 5X^2 + 1$. The top endpoint of U and the constant endpoint of V are both 1, and

$$H_2(U) = 3 \cdot 4 + 1 \cdot 9 = 21, \quad H_2(V) = 1 \cdot 4 + 5 \cdot 9 = 49.$$

Here $\gcd(21, 49) = 7$, so a common divisor does exist and is not small; it is simply coprime to 6.

Corollary 5.5 (no gain from integer scalar content or the stated common divisor at $3/2$). *Under the endpoint hypotheses of Proposition 5.3, neither integer scalar content of specialised rows nor a common divisor of the two specialised evaluations $H_W(U)$ and $H_W(V)$ can supply factors 2 and 3 by those mechanisms in the common-width endpoint architecture studied here.*

Proof. By Theorem 4.1 integer scalar content multiplies the analytic error and the exterior determinant, including the absolute determinant height, by exactly the factors it introduces, so a divisor obtained that way is paid for by the same factor in that height. By Proposition 5.3 an integer dividing both specialised evaluations is divisible by neither 2 nor 3. $\square$

One further consequence of Theorem 5.1 is worth stating, because it bears on the most natural way one might hope to import an existing denominator reduction. Write Φ_m for the m th cyclotomic polynomial and, for coprime $a > b \geq 1$, put $\Phi_m(a, b) = b^{\varphi(m)} \Phi_m(a/b)$ for its homogenisation at the declared width $\varphi(m) = \deg \Phi_m$.

Proposition 5.6 (homogenised cyclotomic values are unit at both endpoints). *Let $a > b \geq 1$ with $\gcd(a, b) = 1$ and let $m \geq 1$. Then $\gcd(\Phi_m(a, b), ab) = 1$. In particular $\gcd(\Phi_m(3, 2), 6) = 1$ for every m .*

Proof. Φ_m is monic and $\Phi_m(0) = \pm 1$, so its coefficients at both declared endpoints are units. Theorem 5.1 applies verbatim with W replaced by $\varphi(m)$: modulo a prime dividing b only the top term of the homogenisation survives, and modulo a prime dividing a only the constant term does. $\square$

The kernel-checked declaration [cyclotomic hom eval is coprime mul](#) proves Proposition 5.6 in the same homogeneous-evaluation representation. It checks Proposition 3.6 under the displayed coprimality assumptions; it does not certify the later analytic deductions or Proposition 8.6.

The methods that reduce denominators at integer bases, the factorial-coset quotients of Rhin and Viola [16] and the order-twelve group and cyclotomic divisor of Zudilin [8, Sec. 4, pp. 159–161], produce their gain as cyclotomic or factorial factors of the coefficient polynomials. Proposition 5.6 says that transporting such a factor through the homogenisation at $(3, 2)$ contributes no power of 2 and no power of 3, whatever its size. This does not make such factors useless: a large odd divisor still reduces Archimedean height, and that is a different account of the same product formula. It does say that

the 2- and 3-primary gain the architecture of Section 10 requires must come from somewhere other than an imported cyclotomic factor. The exclusions therefore reflect the architecture itself, beyond the local behaviour of the two primes involved.

Both ingredients are Lean-checked; the combination is an ordinary deduction and is not separately formalised. The corollary excludes two ways of producing the targeted gain. It does not show that a gain of that kind is necessary for a proof by linear forms at $3/2$.

Corollary 5.5 excludes the two multiplicative mechanisms above, and one candidate pursued in the rest of this section is additive: take an integer combination of several rows and ask that the combination be divisible where the individual rows are not. No single row is multiplied by a scalar. The endpoint congruences are the first case of a divisibility condition that can be imposed to any depth, and it is that condition, read additively, which is counted below.

We first raise the two congruences to prime powers. Fix depths $R, S \geq 0$. For $P \in \mathbb{Z}[X]$ the *bottom jet* $J_{3,R}(P)$ is the residue of $H_W(P)$ modulo 3^R , and the *top jet* $J_{2,S}(P)$ is its residue modulo 2^S ; Theorem 5.1 computes them at $R = S = 1$. Their vanishing is exactly the requested divisibility:

$$J_{3,R}(P) = 0 \iff 3^R \mid H_W(P), \quad J_{2,S}(P) = 0 \iff 2^S \mid H_W(P).$$

These are the checked [bottom-jet divisibility criterion](#) and [top-jet divisibility criterion](#).

The *four-jet signature* of a coefficient pair (U, V) is then the quadruple

$$(J_{3,R}(U), J_{3,R}(V), J_{2,S}(U), J_{2,S}(V)) \in (\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

two residues for each of the two primes, one from each entry of the pair. By the displayed criteria it vanishes exactly when 3^R divides both specialised entries and 2^S divides both.

For this candidate architecture, this turns the targeted local divisor into an additive congruence-kernel problem: what is sought is no longer a common divisor of the two evaluations, but a vector of small integer coefficients on which four residues vanish at once. Since H_W is linear in the coefficients of P , the four-jet signature of a combination is the corresponding combination of signatures, which is what makes the following count possible.

Theorem 5.7 (binary four-jet collision). *Fix a width W and depths R, S , and let $(U_j, V_j)_{j < M}$ be any M pairs of integral polynomials. Call a subset of $\{0, \dots, M-1\}$, equivalently a vector of $\{0, 1\}^M$, a binary selector. If the 2^M binary selectors outnumber the finite four-jet target*

$$(\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

then two distinct subsets have the same four-jet sum. Subtracting their indicator vectors gives a nonzero coefficient vector in $\{-1, 0, 1\}^M$ cancelling all four jets. The target has exact cardinality

$$(3^R)^2(2^S)^2.$$

In particular, if $R > 0$ and $4R + 2S \leq M$, such a collision exists.

Proof. Send each binary selector to the sum of the four-jet signatures it selects. The claimed cardinal inequality and the pigeonhole principle give two distinct selectors in

the same fibre. The cardinality formula is the product of the four cyclic-modulus cardinalities. For $R > 0$,

$$(3^R)^2(2^S)^2 < (4^R)^2(2^S)^2 = 2^{4R+2S} \leq 2^M,$$

which proves the stated sufficient threshold. $\square$

The power bracket improves the generic coefficient $4R$ when the bottom depth is a multiple of 41.

Corollary 5.8 (sharp rank-41 four-jet threshold). *Let $T > 0$. At bottom depth $R = 41T$, any family of $M \geq 130T + 2S$ integral polynomial pairs has two distinct binary selectors with the same four-jet sum. For $T = 1$ the coefficient 130 is exact for this counting argument:*

$$2^{129+2S} < |(\mathbb{Z}/3^{41}\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2|.$$

No exact-optimality assertion is made here for $T > 1$.

Proof. The upper power inequality gives

$$(3^{41T})^2(2^S)^2 < (2^{65})^{2T}(2^S)^2 = 2^{130T+2S} \leq 2^M,$$

so Theorem 5.7 applies. For $T = 1$, direct integer evaluation gives $2^{129} < 3^{82}$; multiplying by $(2^S)^2$ gives the displayed reverse count at $129 + 2S$. $\square$

The sufficient collision threshold is [checked here](#), while its exact unit-block failure one row earlier is [checked here](#).

For all depths at once the ambient-cardinality inequality $2^M > 3^{2R}2^{2S}$ holds exactly when

$$M \geq M_{\text{count}}(R, S) = \lfloor 2R \log_2 3 + 2S \rfloor + 1,$$

because $2R \log_2 3$ is irrational for $R \geq 1$. At $R = 41$ this is $130 + 2S$, and at $R = 41 \cdot 31$ it is $4029 + 2S$, one below the uniform bound $4030 + 2S$ of Corollary 5.8; the corollary trades exactness for a certificate that is a single integer comparison. The name of the parameter 41 is the bottom depth, and the number of input rows is $130 + 2S$.

Counting alone does not ensure that the two selectors produce different analytic remainders. The exact missing step is a bounded-fibre estimate.

Theorem 5.9 (bounded-fibre escape). *Let A and B be finite sets, let $f : A \rightarrow B$, and let $g : A \rightarrow C$ be any map into a set C . Suppose every fibre of g has at most k elements. If*

$$|B|k < |A|,$$

then there exist distinct $x, y \in A$ such that

$$f(x) = f(y) \quad \text{and} \quad g(x) \neq g(y).$$

Thus, with f the four-jet sum and g the analytic remainder, a uniform remainder-multiplicity bound converts surplus selector entropy into a four-jet collision outside the remainder nullspace.

Proof. If every pair in a common f -fibre also had the same g -value, each f -fibre would lie in one g -fibre and hence have size at most k . Summing over the at most $|B|$ fibres of f would give $|A| \leq |B|k$, contrary to the hypothesis. $\square$

This finite escape principle is [checked here](#).

Theorem 5.10 (Bézout–Plücker tail collapse). *Let R_0 be a commutative ring and let $w_n = (A_n, B_n) \in R_0^2$. Suppose that each row is unimodular, meaning that $u_n A_n + v_n B_n = 1$ for some $u_n, v_n \in R_0$, and that every adjacent minor vanishes:*

$$A_n B_{n+1} - B_n A_{n+1} = 0 \quad (n \geq 0).$$

Then every pairwise minor $A_i B_j - B_i A_j$ vanishes. In particular, take $R_0 = \mathbb{Z}/(2^S 3^R)\mathbb{Z}$ with $R > 0$. If $S + 2R \leq k$, there are two distinct binary selectors $s, t \in \{0, 1\}^k$ such that

$$\sum_{i < k} s_i w_i = \sum_{i < k} t_i w_i.$$

Thus the sufficient width is $S + 2R$. The ambient two-coordinate width is $2S + 4R$.

Proof. If (a, b) is unimodular, say $ua + vb = 1$, and $ay - bx = 0$, then

$$(x, y) = (ux + vy)(a, b).$$

Indeed, the first coordinate follows by replacing ay with bx , and the second by the reverse substitution. Apply this identity to consecutive rows: each next row is a scalar multiple of the current one. Induction places the entire tail on the line through w_0 and proves the pairwise-minor assertion. A determinant-one Bézout shear sends w_0 to $(1, 0)$, so all selector sums have only one free residue coordinate and occupy at most $2^S 3^R$ values. Finally

$$2^S 3^R < 2^S 4^R = 2^{S+2R} \leq 2^k,$$

and pigeonhole gives the two selectors. $\square$

No particular coordinate needs to be invertible: modulo six, $(2, 3)$ is unimodular because $-2 + 3 = 1$, although neither entry is a unit. Some nondegeneracy is essential. The three rows $(1, 0), (0, 0), (0, 1)$ have zero adjacent minors but outer minor one; a zero middle row transmits no information between its neighbours.

Lean checks the unit-coordinate form inside the supported root: the [adjacent-to-pairwise determinant propagation](#), the resulting [modular selector collision](#), and the [explicit \$S+2R\$ threshold](#). Each of the three assumes that the second coordinate of every row is a unit. The rowwise-coprime statement proved above is the stronger one. This conditional theorem is stronger than the ambient four-jet count only after its minor-vanishing hypothesis has been established. No such all-tail hypothesis is proved here for an actual q -Apéry or Zudilin family, and the theorem says nothing about whether the resulting selector difference has nonzero analytic remainder.

Boundary. Corollary 5.8 is a sharp finite kernel statement at $T = 1$, not an analytic nonvanishing theorem. Theorem 5.9 identifies the precise extra input needed to escape

the nullspace, but this paper does not prove a multiplicity bound for the actual q -Apéry or Zudilin remainder family.

The target count is the checked [four-jet target cardinality](#); the abstract collision is the checked [four-jet pigeonhole kernel](#), and the linear sufficient condition is the checked [rank–depth collision threshold](#). The count is a routine pigeonhole; the reformulation above is what makes it relevant. Pigeonhole cancellation itself requires no independence. Additional information about the input family is needed to ensure that the resulting nonzero selector difference has a nonzero combined polynomial pair and analytic remainder. None of the statements proved here supplies such a family or proves either nonvanishing conclusion.

Example 5.11. At depths $R = S = 1$ the four-jet target is $(\mathbb{Z}/3\mathbb{Z})^2 \times (\mathbb{Z}/2\mathbb{Z})^2$, of cardinality $9 \cdot 4 = 36$, and the threshold reads $M \geq 4 \cdot 1 + 2 \cdot 1 = 6$. With six pairs there are $2^6 = 64$ binary selectors against 36 targets, so two of them collide and their difference is a vector in $\{-1, 0, 1\}^6$, not identically zero, killing all four jets.

Informally, the theorem says only this: once there are at least $4R + 2S$ pairs and the bottom depth R is positive, some coefficient vector in $\{-1, 0, 1\}^M$, not identically zero, kills all four jets of the corresponding combination. It does not say that the combination is nonzero as a pair of polynomials, and it does not say that its remainder is nonzero. Those are the two obligations Problem 10.1 carries.

One further exclusion is recorded here. Its subject is the scalar parameters of Zudilin’s cone, and it says nothing about the coefficient polynomials.

Theorem 5.12 (scalar margin no-go). *Let $C_1 > 0$. If $C_0 \leq 0$ or $2C_0 \leq C_1$, then*

$$C_0 \log 3 - C_1 \log 2 < 0.$$

In the positive branch $C_0 > 0$ and $2C_0 \leq C_1$, the stronger estimate is

$$C_0 \log 3 - C_1 \log 2 < -\frac{17}{41} C_0 \log 2.$$

Written multiplicatively, the conclusion is $3^{C_0} < 2^{C_1}$. The inequality is immediate from $\log 3 < 2 \log 2$ and $C_1 > 0$, and is the checked [three-halves scalar margin](#). The positive-branch deficit is the checked [17/41 margin](#). The interest is in the fence around it. The primary Zudilin theorem [8, Theorem 1, p. 154; Sec. 5, pp. 161–162] supplies an integer-base irrationality-exponent estimate on its parameter cone, and the elementary inequality $\mu \geq 2$ then forces $2C_0 \leq C_1$ whenever $C_0 > 0$; Lean checks that implication separately. The primary 2004 theorem is stated for an integer inverse base. The rational specialisation proved earlier in this record has separate source-applied declarations. The scalar statement in this paragraph is only the displayed inequality; it neither constructs a new family at $3/2$ nor formalises a universal exclusion of linear-form methods.

6 Failure of coordinatewise clearing at $3/2$

This section and the next return to the elementary route and record what the residue s^n costs there. We first isolate the arithmetic that the clearing scheme leaves behind, in a

form that does not mention the series. The point of doing so is that the clearing argument asks for two things at once, that a power of the numerator divide what has been accumulated and that what survives be small, and these are easier to play off against each other once the series has been discarded and only six natural numbers remain.

Definition 6.1 (coordinatewise corridor). Let a, b, N, K, Q, D be natural numbers. Say that (a, b, N, K, Q, D) is a *coordinatewise corridor* when

$$a > 0, \quad Q > 0, \quad D > 0, \quad D \leq N + K, \quad a^K \mid QD, \quad Qb^{N+K+1} < a^{K+1}.$$

The reading is: a and b are the numerator and denominator of the base, playing the roles of r and s in the introduction, so that $(a, b) = (3, 2)$ is the case of interest; N is the shift, K is the width of the cleared window, Q is the accumulated clearing factor, and D is the final coefficient being cleared. The bound $D \leq N + K$ is the only property of the coefficient used; for the divisor-counting coefficient it holds because $\tau(n) \leq n$. The divisibility $a^K \mid QD$ is the requirement that clearing succeeded coordinatewise, and the last inequality is the tail estimate that makes the trapped integer smaller than 1. The name records the shape of the constraint: the divisibility bounds a^K from above by $Q(N + K)$, the tail estimate bounds a^{K+1} from below by Qb^{N+K+1} , and admissible parameters must fit in the band between them. The definition is the [corridor predicate](#).

Theorem 6.2 (corridor bound). *If (a, b, N, K, Q, D) is a coordinatewise corridor, then*

$$b^{N+K+1} < a(N + K).$$

Proof. A routine divisibility computation. Since $QD > 0$ and $a^K \mid QD$, we have $a^K \leq QD$, and $D \leq N + K$ gives $a^K \leq Q(N + K)$. Hence

$$Qb^{N+K+1} < a^{K+1} = a^K \cdot a \leq Q(N + K) \cdot a,$$

and cancelling the positive factor Q gives the claim. □

Formalised as the [power-versus-linear consequence](#).

The inequality of Theorem 6.2 is where the integer and rational cases part. At $b = 1$ it reads $1 < a(N + K)$, which holds for every $a \geq 2$ and every nonempty window; this necessary inequality imposes no obstruction. The other corridor hypotheses remain in force. At $b \geq 2$ the left side is exponential in $N + K$ and the right side is linear, so the corridor can survive only for small $N + K$. At $(a, b) = (3, 2)$ the crossing has already happened at the smallest admissible window.

Example 6.3. Take the smallest window, $N = K = 1$, and numerator $a = 3$. At $b = 1$ the tuple $(3, 1, 1, 1, 3, 1)$ is a corridor: $D = 1 \leq 2$, the divisibility reads $3 \mid 3$, and the tail estimate reads $3 \cdot 1^3 = 3 < 3^2 = 9$. At $b = 2$ no choice works. The tail estimate becomes $Q \cdot 2^3 < 3^2$, which forces $Q = 1$; the divisibility then reads $3 \mid D$, and the only candidates are $D = 1$ and $D = 2$, neither divisible by 3. This is the proof of Theorem 6.2 in miniature: it derives $a^K \leq Q(N + K)$, here $3 \leq 2$.

Proposition 6.4. *For every natural number $x \geq 2$ we have $3x < 2^{x+1}$.*

Proof. A routine induction from $x = 2$, where $6 < 8$. For the step, $2^{x+1} \geq 2^2 > 3$ when $x \geq 1$, so $3(x+1) = 3x+3 < 2^{x+1} + 2^{x+1} = 2^{x+2}$. $\square$

Formalised as the [exponential comparison](#).

Theorem 6.5 (no corridor at base $3/2$). *For all $N \geq 1$ and $K \geq 1$ and all natural Q, D , the tuple $(3, 2, N, K, Q, D)$ is not a coordinatewise corridor.*

Proof. A corridor would give $2^{N+K+1} < 3(N+K)$ by Theorem 6.2, contradicting Proposition 6.4 applied to $x = N+K \geq 2$. $\square$

Formalised as the [corridor exclusion at three halves](#).

Theorem 6.5 excludes the coordinatewise clearing scheme at $3/2$, and nothing else: it does not bound the denominator of $F(3/2)$, it does not show that $F(3/2)$ is irrational, and it does not show that $F(3/2)$ is rational. It also does not cover a clearing scheme of a different shape, since the corridor fixes one divisibility pattern and one tail inequality.

7 The cleared-tail recurrence and the size of the forcing term

Theorem 6.5 says that one scheme fails. This section identifies the quantity responsible, as an exact recurrence.

Let r, s, B, F be rationals with $r \neq 0$ and let $c : \mathbb{N} \rightarrow \mathbb{Q}$ be arbitrary. Define the prefix and the cleared tail state by

$$P_N = \sum_{m=0}^{N-1} \frac{c(m+1)s^{m+1}}{r^{m+1}}, \quad U_N = Br^N(F - P_N). \quad (*)$$

Thus P_N is the partial sum of $\sum_{n \geq 1} c(n)(s/r)^n$ through level N , and U_N is the tail of a putative value F after that level, scaled by Br^N . These are the [rational-base prefix](#) and the [cleared tail state](#).

Theorem 7.1 (cleared-tail recurrence). *Let $r, s, B, F \in \mathbb{Q}$ with $r \neq 0$, let $c : \mathbb{N} \rightarrow \mathbb{Q}$, and let P_N and U_N be as in $(*)$. Then for every N ,*

$$U_{N+1} = rU_N - Bc(N+1)s^{N+1}.$$

Proof. An immediate computation. Expanding $P_{N+1} = P_N + c(N+1)s^{N+1}/r^{N+1}$ and $r^{N+1} = r^N \cdot r$ in the definition of U_{N+1} and clearing the denominator r^{N+1} , which is nonzero, gives the identity. $\square$

Formalised as the [cleared-tail recurrence](#).

The recurrence has a linear part rU_N and a forcing term $Bc(N+1)s^{N+1}$, the inhomogeneous term the state receives at each step. The direct-clearing route studied here works only if the state remains in a bounded window, and the forcing term is what that window must absorb. Its size is what separates the two denominator regimes.

Theorem 7.2 (the forcing term). *Let s, B be natural numbers and $c : \mathbb{N} \rightarrow \mathbb{N}$, and put $G_N = Bc(N+1)s^{N+1}$.*

1. If $s \geq 2$, $B \geq 1$ and $c(N+1) \geq 1$, then $2^{N+1} \leq G_N$.
2. If $s = 1$, then $G_N = Bc(N+1)$.

Proof. Both parts are routine. For the first, $2^{N+1} \leq s^{N+1} = 1 \cdot s^{N+1} \leq Bc(N+1)s^{N+1}$, using $Bc(N+1) \geq 1$. The second part is the definition with $s = 1$. $\square$

Formalised as the [exponential lower bound](#) and the [integer-base collapse](#).

At $s = 1$ the forcing term is $Bc(N+1)$, so it grows only as fast as the coefficient; for the divisor function that is $O(N^\varepsilon)$ for every $\varepsilon > 0$, and a bounded-state argument has room. At $s \geq 2$ the same term is at least 2^{N+1} whenever the coefficient is nonzero.

Example 7.3. Take $B = 1$, $c = \tau$ and $N = 9$, so that the coefficient is $\tau(10) = 4$. At $s = 1$ the forcing term is 4. At $s = 2$ it is $4 \cdot 2^{10} = 4096$, and part (1) of Theorem 7.2 already guarantees at least $2^{10} = 1024$ without knowing the coefficient at all.

This is an exact lower bound on one quantity, and it is all that is proved. It is not a proof that no bounded-state argument exists at $s \geq 2$; the theorem that one particular scheme fails is Theorem 6.5, and the bound here records the size of the term that scheme would have to absorb.

8 The height criterion at 7/2

In 1994 Bundschuh and Väänänen proved an irrationality criterion for a family of rational bases cut out by a height condition [6, Theorem 2, p. 177; hypotheses pp. 175–176]. We keep their notation: q is the base, and α and λ are the parameters of the criterion. In its special case $\alpha = -1$, the printed hypothesis is

$$\lambda < \left(\frac{1}{2} + \frac{1}{\pi^2} \right)^{-1}.$$

At $q = 7/2$ the Archimedean parameter is $\lambda = \log 7 / \log(7/2)$. The criterion therefore applies once the following strict inequality is checked.

Theorem 8.1 (the 7/2 height condition).

$$\frac{\log 7}{\log(7/2)} < \left(\frac{1}{2} + \frac{1}{\pi^2} \right)^{-1}.$$

Numerically the two sides are 1.5533 ... and 1.6630 ..., so the condition holds with a margin of about 0.11. The Lean proof factors the estimate through the explicit [height-region predicate](#) and the [integer certificate](#) $2^{18} < 7^7$, the resulting [logarithmic ratio bound](#) $\log 2 / \log 7 < 7/18$, the [\$\pi\$ -bound](#) $1/\pi^2 < 1/9$, and the [strict margin](#)

$$\frac{\log 2}{\log 7} < \frac{1}{2} - \frac{1}{\pi^2}.$$

The [final height inequality](#) then rewrites $\log(7/2) = \log 7 - \log 2$ and closes the displayed condition.

This formalises the complete elementary parameter check at $q = 7/2$; it does not formalise Bundschuh and Väänänen’s analytic irrationality theorem, whose proof occupies pp. 189–193 of the source. The conclusion that $F(7/2)$ is irrational is consequently cited from that theorem, not claimed as a Lean theorem here.

The criterion does not cover $3/2$, and the results of Sections 6 and 7 give no evidence either way about whether $F(3/2)$ is irrational.

8.1 The 81/200 logarithmic region

Consider the rational-height region

$$\frac{\log b}{\log a} < \frac{81}{200} \quad (a > b > 0).$$

The rational number $81/200$ is not itself a threshold with an analytic theorem attached to it. It is the lower half of the rational bracket $81/200 < \theta^* < 1/2$ of Section 2.5 around the derived constant of Theorem 2.2, and its role is to make membership of the region of that theorem decidable by an integer comparison. The Lean module of this library treats the displayed inequality as a definition and proves elementary memberships and exclusions. In particular,

$$\frac{2}{5} < \frac{\log 4}{\log 31} < \frac{81}{200} < \frac{\log 2}{\log 3}.$$

The first two comparisons come respectively from $31^2 < 4^5$ and $4^{200} < 31^{81}$; the last comes from $3^{81} < 2^{200}$. The lower bound is the checked theorem [two-fifths lower bound](#). Consequently $31/4$, and every positive power $(31/4)^r$, lies in the enlarged region ([checked, power family](#)). The same base lies strictly outside the earlier Bundschuh–Väänänen region ([checked](#)), so the $81/200$ inequality defines a strict set-theoretic enlargement of the earlier recorded logarithmic region. Combined with the bracket $81/200 < \theta^*$, these memberships are exactly the finite input to Theorem 2.3, which is where the irrationality of $F(31/4)$ and of every $F((31/4)^r)$ is proved.

It still does not approach $3/2$. The exact comparison

$$3^{81} < 2^{200} \quad \Rightarrow \quad \frac{81}{200} < \frac{\log 2}{\log 3}$$

is Lean-checked, as is the conclusion that $3/2$ belongs to neither height region ([boundary, exclusion](#)). Thus $81/200$ is the larger of the two explicitly defined cutoffs used below, while a cutoff that includes $3/2$ must be strictly larger than $\log 2 / \log 3 \approx 0.6309$.

9 A denominator-exponent model for Padé-type forms

A second external route builds the linear forms of Section 4 by Padé approximation. Such a construction produces its coefficients as sums of rational summands, and to obtain integer rows one multiplies through by a single common denominator. The book-keeping obligation is then to check that no summand needs a larger denominator than the one proposed, which is a comparison between the exponents alone. This section does that comparison, and only that.

No source construction is named here for the exponent expressions below, and they are checked as displayed polynomial identities. No derivation of them is given here. Giving them Padé-theoretic force would additionally require deriving the expressions from a stated homogenised coefficient formula, proving integrality of the coefficients after multiplication by the proposed common denominator, and establishing nonvanishing and decay of the associated remainder. None of those three is claimed here.

In the model, the common denominator exponent is $E_n = (3n^2 - n)/2$. Since only doubled exponents occur below, every statement lives over $\mathbb{Z}$ and no parity bookkeeping is needed; we write $\tilde{E}_n = 2E_n = 3n^2 - n$.

Proposition 9.1 (exponent model: summand bound and exact gap). *Let $\tilde{E}_n = 3n^2 - n$ and put*

$$\tilde{P}(n, k) = 2(k(n - k) + nk) + k(k - 1),$$

$$\tilde{Q}(n, m) = 2(n^2 - n) + j^2 + 2jm + j - m^2 + 3m, \quad j = n - m - 1.$$

Then, for integers n, k, m :

1. if $0 \leq k \leq n$, then $\tilde{P}(n, k) \leq \tilde{E}_n$, and the gap factors as $\tilde{E}_n - \tilde{P}(n, k) = (n - k)(3n - k - 1)$;
2. $\tilde{E}_n - \tilde{Q}(n, m) = 2(n + m(m - 1))$ identically;
3. if $n \geq 0$ and $m \geq 1$, then $\tilde{Q}(n, m) \leq \tilde{E}_n$.

Part (1) is the [summand exponent bound](#), part (2) the [exact gap identity](#), and part (3) the [maximal exponent bound](#). The gap in (2) is an identity in $\mathbb{Z}[n, m]$, so (3) follows from $m(m - 1) \geq 0$ and $n \geq 0$; the hypothesis $m \geq 1$ names the range in which the corresponding summand does not vanish, and it is not the sharpest hypothesis under which the inequality holds.

Example 9.2. At $n = 2$ the proposed doubled exponent is $\tilde{E}_2 = 10$, and $\tilde{P}(2, k)$ takes the values 0, 6, 10 at $k = 0, 1, 2$. The three gaps are 10, 4, 0, matching the factorisation $(2 - k)(5 - k)$ of part (1); the summand at $k = 2$ is the one that saturates the proposed denominator. For part (2), at $m = 1$ we have $j = 0$ and $\tilde{Q}(2, 1) = 6$, with gap $4 = 2(2 + 1 \cdot 0)$.

These are routine inequalities between polynomials in the exponents. They establish that the proposed exponent $\tilde{E}_n$ dominates the two displayed summand exponent expressions, and nothing further. Positivity of the remainder, its rate of decay, and the comparison of that rate against the denominator height are the analytic obligations, and none of them is treated here, so nothing in this section is an irrationality measure.

10 Complements and further questions

The universal assertion of Problem #1049 is not settled here. Theorem 5.7 suggests the following precise sufficient subproblem for one common-width additive architecture. It is not asserted to be necessary for every proof of irrationality.

Problem 10.1 (common-width simultaneous endpoint-jet construction). Exhibit an integer constant $C \geq 1$ and, for every sufficiently large positive integer n , positive integers W_n, R_n, S_n, M_n such that

$$n^2 \leq W_n, R_n, S_n \leq Cn^2, \quad 4R_n + 2S_n \leq M_n \leq Cn^2,$$

together with polynomial pairs $(U_{n,j}, V_{n,j}) \in \mathbb{Z}[X]^2$ for $0 \leq j < M_n$, each of degree at most the common declared width W_n , whose specialised integer rows are primitive:

$$\gcd(H_{W_n}(U_{n,j}), H_{W_n}(V_{n,j})) = 1.$$

Find a nonzero vector $\lambda^{(n)} \in \{-1, 0, 1\}^{M_n}$ for which, on putting

$$U_n = \sum_{j < M_n} \lambda_j^{(n)} U_{n,j}, \quad V_n = \sum_{j < M_n} \lambda_j^{(n)} V_{n,j},$$

the pair (U_n, V_n) is not $(0, 0)$, all four common-width jets vanish,

$$J_{3,R_n}(U_n) = J_{3,R_n}(V_n) = 0, \quad J_{2,S_n}(U_n) = J_{2,S_n}(V_n) = 0,$$

where every jet in this display is formed using the declared width W_n , and the resulting divided integer linear form

$$A_n = \frac{H_{W_n}(U_n)}{3^{R_n} 2^{S_n}}, \quad B_n = \frac{H_{W_n}(V_n)}{3^{R_n} 2^{S_n}}, \quad \rho_n = A_n F(3/2) - B_n$$

satisfies the explicit analytic condition

$$0 < |\rho_n| < \frac{1}{n}.$$

The jet equations make A_n, B_n integers. A solution would prove irrationality: if $F(3/2) = a/b$ in lowest terms, every nonzero ρ_n has absolute value at least $1/b$, contradicting the displayed bound for $n > b$. Theorem 5.7 supplies only a nonzero signed relation with the four jet equations once the pairs and size inequality are present; it does not supply primitive input rows, a nonzero combined polynomial pair, or the nonvanishing and decay of ρ_n .

Integer rescaling and a common divisor of the two specialised evaluations are the two mechanisms excluded by Corollary 5.5. Polynomial factors before specialisation, cross-row or determinant-specific arithmetic, cyclotomic factors, other additive constructions, and entirely different architectures remain outside that corollary and are not decided either way.

The remaining linear-form argument therefore has two independent gates: first find a four-jet collision that does not collapse algebraically or analytically; only then ask whether its divisibility and decay beat its height. We state those gates separately below.

10.1 Exact kernel escape

Fix, for each n , a declared width W_n and a specified family

$$(U_{n,j}, V_{n,j}, \mathcal{R}_{n,j})_{j < M_n},$$

where $U_{n,j}, V_{n,j} \in \mathbb{Z}[X]$ and $\mathcal{R}_{n,j}$ is the corresponding remainder function. Normalisation must be fixed before the jet map is formed. Call the family *polynomially primitive* when the common coefficient content of the pair in $\mathbb{Z}[X]^2$ is divided out before specialisation; as the Terminology paragraph records, that is a different operation from primitive normalisation of a specialised integer row, which is what Problem 10.1 imposes.

The specialised row content

$$g_{n,j}^{\text{spec}} = \gcd(H_{W_n}(U_{n,j}), H_{W_n}(V_{n,j}))$$

and the content of a final additive combination are separate quantities; they are not silently divided out. If either is divided out later, both the height and the remainder below are measured after that same division. Under the unit endpoint hypotheses, $g_{n,j}^{\text{spec}}$ is coprime to 6, so such division does not manufacture or destroy the 2- and 3-primary jet vanishing.

Dividing a specialised row by its content changes the polynomial family it came from, and lifting the divided row back is a constrained problem. At width W the map $P \mapsto H_W(P)$ on integer polynomials of degree at most W is surjective onto $\mathbb{Z}$, since 3^W and 2^W are coprime, so a lift always exists; what is not automatic is a lift carrying the intended remainder identity and a uniform coefficient-height bound. At width 1, for instance, $(X+1, X+1)$ specialises to $(5, 5)$, whose primitive normalisation $(1, 1)$ lifts to $(X-1, X-1)$ and not to any rescaling of the original pair. Any lift used below is therefore supplied together with its width, its height bound and its exact remainder.

For target depths R_n, S_n , let $J_n(\lambda)$ be the four-jet signature of the pair $\sum_j \lambda_j (U_{n,j}, V_{n,j})$, and define

$$\begin{aligned} C_n &= \{\lambda \in \{-1, 0, 1\}^{M_n} \setminus \{0\} : J_n(\lambda) = 0\}, \\ K_n^{\text{poly}} &= \left\{ \lambda \in \{-1, 0, 1\}^{M_n} : \sum_j \lambda_j U_{n,j} = 0, \sum_j \lambda_j V_{n,j} = 0 \right\}, \\ K_n^{\text{rem}} &= \left\{ \lambda \in \{-1, 0, 1\}^{M_n} : \sum_j \lambda_j \mathcal{R}_{n,j}(3/2) = 0 \right\}. \end{aligned}$$

The second nullspace concerns the value at $3/2$; an identically zero remainder function is a still stronger collapse and is automatically bad.

The exact pigeonhole condition is

$$2^{M_n} > 3^{2R_n} 2^{2S_n}, \quad \text{equivalently} \quad M_n > 2R_n \log_2 3 + 2S_n. \quad (8.1)$$

Thus the least admissible integer rank is

$$M_{\min}(R, S) = \lfloor 2R \log_2 3 + 2S \rfloor + 1.$$

The checked condition $M \geq 4R + 2S$ for $R > 0$ is a convenient sufficient corollary, not the exact threshold. Moreover, with $N = 2^{M_n}$ and $Q = 3^{2R_n} 2^{2S_n}$, convexity of the fibre sizes gives at least

$$\frac{1}{2} \left(\frac{N^2}{Q} - N \right) \quad (8.2)$$

unordered equal-signature selector pairs. A normality estimate can therefore win by showing that fewer than this many collision pairs land in the two bad nullspaces.

Problem 10.2 (four-jet kernel escape). For one literal polynomially primitive family satisfying (8.1), prove, preferably by comparing the lower bound (8.2) with the bad-pair multiplicities, that

$$C_n \setminus (K_n^{\text{poly}} \cup K_n^{\text{rem}}) \neq \emptyset$$

for all sufficiently large n .

Theorem 5.7 supplies only $C_n \neq \emptyset$. It gives a nonzero selector difference, but it gives neither polynomial-pair nonvanishing nor remainder nonvanishing. Problem 10.2 is finite algebra and normality; it makes no asymptotic product-formula claim. It is the first of the two gates in Problem 10.1: it asks for the two nonvanishing conclusions, and not for the analytic condition stated there.

The specialised lattice is a separate calculation. Let primitive integer rows $u_j \in \mathbb{Z}^2$ span a rank-two lattice L , and let g be the gcd of their 2×2 minors. Since at least one row is primitive, the Smith invariants are $1, g$ [32, Thms. 2.3–2.4, p. 3]. Hence, for $D \geq 1$,

$$|\text{im}(L \rightarrow (\mathbb{Z}/D\mathbb{Z})^2)| = \frac{D^2}{\gcd(g, D)}, \quad [\mathbb{Z}^2 : (L \cap D\mathbb{Z}^2)/D] = \frac{g}{\gcd(g, D)}.$$

Indeed an integral unimodular change of coordinates takes L to $\mathbb{Z} \times g\mathbb{Z}$; reduction modulo D and intersection with $D\mathbb{Z}^2$ give the two formulas. This is Smith normal form over $\mathbb{Z}$, not over $\mathbb{Z}[p]$. It measures the actual image, rather than the ambient four-jet target. If two independent divided rows have coordinate height at most H and remainders of absolute value at most ε , their nonzero integer determinant gives the necessary inequality $2H\varepsilon \geq g / \gcd(g, D)$. That is not a necessary condition for producing a single nonzero form. If the specialised rows are not primitive, let $d_1 \mid d_2$ be the positive Smith invariants instead. For $D \geq 1$ the corresponding formulas are

$$|\text{im}(L \bmod D)| = \frac{D^2}{\gcd(d_1, D) \gcd(d_2, D)}, \quad [\mathbb{Z}^2 : (L \cap D\mathbb{Z}^2)/D] = \frac{d_1 d_2}{\gcd(d_1, D) \gcd(d_2, D)}.$$

They follow by applying the one-dimensional calculation to each summand $d_i\mathbb{Z}$. Polynomial coefficient content alone does not determine either integer Smith invariant after specialisation.

10.2 Determinant families: definition before conjecture

The earlier exploratory record mentions a contiguous a_0 -shift matrix at rank $13n + 2$, but does not give its entries, its width or its relation to the number of coefficient pairs.

Consequently the assertion $\det A_{n,13n+2} \neq 0$ is not yet a well-defined public conjecture. Before further computation, a family specification must fix the entrywise formula, allowed shifts, normalisation, width and remainder identity. Only then is it meaningful to test the first nonzero minor, row contents and jets, or to ask whether adding one a_1 -shift increases rank. The earlier proposed questions are retained in the archival edit record, not presented as if their missing matrix were already defined.

10.3 Asymptotic adequacy after kernel escape

The next displayed margin is an additional research target for a specified height-remainder architecture, not a necessary irrationality criterion. First fix one such architecture and define its height $H_n > 0$, undivided nonzero remainder L_n and exactly once-counted certified divisor $D_n = 3^{R_n} 2^{S_n}$. If another divisor is used, replace the two logarithmic terms below by $\log D_n$. A scalar-form height and an exterior-determinant height cannot be interchanged: the comparison with a nonzero integer must be derived for the actual objects selected. For a scalar construction, integral coefficients and $0 < |L_n|/D_n \rightarrow 0$ already suffice, with no extra height factor.

Problem 10.3 (negative normalised product-formula margin). Prove the explicit estimate

$$\limsup_{n \rightarrow \infty} \frac{\log H_n + \log |L_n| - R_n \log 3 - S_n \log 2}{n^2} < 0. \quad (8.3)$$

Every denominator, row content and final-combination content must already be included in H_n and L_n .

Nonvanishing alone does not address (8.3). Conversely, excellent formal decay is irrelevant if every jet collision lies in a bad nullspace. An earlier exploratory adjacent-exterior calculation reports a positive normalised exponent of about $110.850 n^2$. It is not a checked theorem or a reproduced certificate of this revision, and it supplies no rigorous numerical target until that architecture and estimate are established. A proposed divisor saving must be compared with the height and remainder of the same explicitly normalised objects.

10.4 One exact alternative-criterion test

A separate possible method is Mahler's method, and its first applicability test has a negative answer. Bell and Smertnig's classification shows that the divisor generating series $\mathcal{L}$ below is not k -Mahler for any $k \geq 2$ [26, Thm. 1.3 and the consequences on p. 3]. The proposition below proves the simultaneous $2/3$ case directly from theorems of Adamczewski and Bell and of Rivin. The 2026 works are cited as the identified preprints, not as journal publications. A function-level obstruction to these functional equations does not determine the arithmetic nature of any one rational-base value.

Proposition 10.4 (no finite simultaneous $2/3$ -system). *Let*

$$\mathcal{L}(z) = \sum_{n \geq 1} \frac{z^n}{1 - z^n}.$$

There is no finite-dimensional $\mathbb{Q}(z)$ -vector space that contains $\mathcal{L}$ and is stable under both $z \mapsto z^2$ and $z \mapsto z^3$.

Proof. Suppose V were such a space, of dimension d . Stability under $z \mapsto z^2$ places the $d+1$ elements $\mathcal{L}(z), \mathcal{L}(z^2), \dots, \mathcal{L}(z^{2^d})$ in V , so they are linearly dependent over $\mathbb{Q}(z)$; clearing denominators gives polynomials $P_0, \dots, P_d$, not all zero, with $\sum_{i=0}^d P_i(z) \mathcal{L}(z^{2^i}) = 0$. Thus $\mathcal{L}$ is 2-Mahler, and the same argument under $z \mapsto z^3$ makes it 3-Mahler. Since 2 and 3 are multiplicatively independent, a theorem of Adamczewski and Bell [15, Thm. 1.1, p. 6] then forces $\mathcal{L}$ to be a rational function. That contradicts Rivin's periodic-coefficient corollary [23, Cor. 6.4, p. 9], by which $\mathcal{L}$ is not rational. $\square$

The proposition uses no property of the point $2/3$: the obstruction is functional and appears before regularity at a particular point is considered. For this scalar function the single-base classification of Bell and Smertnig contains the proposition; the proof above isolates simultaneous closure. A construction using additional functions or functional relations must specify those functions and its closure conditions; the single-base statement is not an obstruction to every approximation method. The classification and the two external ingredients in the proposition are cited, not proved here.

10.5 A limitation of the rectangular exponent model

The module `HermitePadeNoGo` defines an explicit rectangular two-parameter exponent model. Writing $\sigma = 1 + \rho + u$, its denominator-cleared gap has the exact expansion

$$-\pi^2 \rho^2 - \pi^2 \rho u - 2\pi^2 \rho - 2\rho^2 - 10\rho u - 4\rho - 6u^2 - 8u.$$

Thus for $\rho \geq 0$ and $\sigma \geq 1 + \rho$ the cleared gap is nonpositive, and it vanishes exactly when $\rho = 0$ and $\sigma = 1$ ([exact expansion](#), [nonpositivity](#), [equality case](#)). Equivalently, within that model the displayed threshold never exceeds the classical one-function margin, with equality only at the classical endpoint ([bound](#), [equality](#)). This is an algebraic theorem about the four defined exponent expressions and only that explicit rectangular model. It does not construct polynomials, remainders, integrality, a determinant, or asymptotics, and it is not a method-universal no-go theorem. The separate $81/200$ cutoff of the preceding section satisfies

$$\frac{81}{200} < \frac{\log 2}{\log 3}.$$

Thus “improve the height theorem” has a precise numerical target.

Problem 10.5 (optimal admissible height threshold). Specify a concrete class $\mathcal{A}$ of primitive, noncollapsed constructions, excluding scalar row rescaling, proportional permutation-orbit forms, pure row-content gain, and constructions governed by the rectangular exponent model above. For that class define

$$\theta_* = \sup_{\alpha \in \mathcal{A}} \frac{C_0(\alpha)}{C_1(\alpha)}.$$

A useful result would be an explicit admissible construction with $\theta_* > \theta^*$, an exact value of θ_* , or a converse bound $\theta_* < \log 2 / \log 3$. Another possibility is a class theorem

reducing all members to one of the specified scalar or rectangular mechanisms. Here θ^* is the exactly defined cutoff of Theorem 2.2; no new rounded threshold is intended.

Reaching $3/2$ requires a threshold strictly beyond $\log 2 / \log 3 \approx 0.6309$; exceeding $81/200 = 0.405$ is insufficient. The unrestricted question of which rational bases give irrational values is not settled here and is not reduced to any one of these problems.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. Proof authority rests in those pinned sources. The present text is exposition and navigation.

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A Guide to the formal sources

The public source snapshot used for this editorial audit is `3d6d938d696fed0fb71dd55115a18a73738ff223`; the parallel release snapshot is `52f29ad173b04e3bac941b3663f2b9aeb5de0bb`. The supplied index distinguishes public CI-checked declarations, release-only declarations and declarations outside the checked build. Those categories must not be conflated.

The main source-applied irrationality and measure endpoints are in `Erdos1049/PaperR17/SourceConsumers.lean`. The file constructs `actual_cancelled_supply` before applying it in `rational_base_region`, `thirtyone_four_powers` and `rational_base_power_measure`. The all-rank formal-order endpoints are in `Erdos1049/AllRow/Producer.lean`; the producer proves the actual source-row initial monomials and applies the determinant consumer. In particular these are not merely interfaces assuming the missing rows.

Other modules check corridor and recurrence identities, degree bookkeeping, endpoint congruences, finite collision principles and specified exponent models. The stronger ordinary unimodular-row Bézout–Plücker statement must be distinguished from the checked unit-coordinate version. The finite coefficient-moment and polynomial content calculations in Section 3.2 are exact computer algebra, not Lean declarations.

The accompanying audit reports the packet’s CI evidence and a static inspection of the supplied sources. It is not a fresh kernel run or an axiom audit. Canonical links should be checked by declaration name as well as line number when this text is integrated; historical linked revisions are retained where no same-declaration mapping was established.

Four-jet collision family. The ordinary proof uses rowwise unimodularity and zero adjacent minors. The linked Lean statements instead assume that each second coordinate is a unit. The example $(2, 3)$ modulo six satisfies the former condition but not

the latter. Thus the ordinary generalisation is not silently included in the scope of the checked unit-coordinate statements. Neither version establishes the required hypotheses for an actual all-tail approximation family, or proves nonzero analytic remainder for a selector difference.

B Result registry and publication scope

The project registry records editorial classifications and result-atom coordinates. These are retained in the accompanying archival catalogue, not used as mathematical hypotheses or as a substitute for the theorem statements in this paper. In particular, registry promotion status and comparator selection are distinct from mathematical strength and from the scope of a Lean build. The principal arguments are indexed in the reading map and the formal-source appendix above.

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